

Topology

Summary with my Intuitions

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1 Definition of Topology

In many times, we ‘draw’ some mathematical objects especially sets. One of the key intuition of studying topology is how can we ‘draw’ such things. To draw somethings formally, we need to define bunch of things like *shape of sets*. As we describe *shape of groups* with its inner structures like subgroups, we describe *shape of sets* with its inner structures like subsets.

Definition 1.1. Topology

A topology $\mathcal{T}$ on a set X is collection of subsets of X having following axioms.

1. $\emptyset, X \in \mathcal{T}$
2. $\mathcal{T}$ is closed under arbitrary unions.
3. $\mathcal{T}$ is closed under finite intersections.

Then we denote the pair $(X, \mathcal{T})$ is *topological space* (or just *space*) and call every $U \in \mathcal{T}$ by is *open set* in X . Then now X is shaped by $\mathcal{T}$ with certain axioms such that related subsets of X . This is very similar to defining groups.

Definition 1.2. Group

Let G be a set and $\star : G \times G \rightarrow G$ is binary operation. $(G, \star)$ is group when have following axioms.

1. $\forall a, b, c \in G, \quad a \star (b \star c) = (a \star b) \star c.$
2. $\exists 1_G \in G$ such that $\forall x \in G, 1_G \star x = x \star 1_G = x.$
3. $\forall x \in G, \exists x^{-1} \in G$ such that $x \star x^{-1} = x^{-1} \star x = 1_G.$

Well, the definition of topology is quite less intuitive. However we already familiar with ‘shapes’ in $\mathbb{R}^n$ and often draw them in natural. Topology is just generalize such notions into general sets. In analysis in $\mathbb{R}^n$, the key block was *open balls*. We generalize this notion into general set also, which is called *basis*.

Definition 1.3. Basis

A *basis* $\mathcal{B}$ is collection of subsets of X having following axioms.

1. $\forall x \in X, \exists B \in \mathcal{B}$ such that $x \in B.$
2. $\forall B_1, B_2 \in \mathcal{B},$ if $x \in B_1 \cap B_2, \exists B^* \in \mathcal{B}$ such that $x \in B^* \subseteq B_1 \cap B_2.$

This is very natural property when we imagine open balls in $\mathbb{R}^n$. Another good

motivation to understand this is that, understand each $B \in \mathcal{B}$ is collection of ‘near’ elements. In metric space we’ll define later, such notion of distance is very natural but here, we need to understand abstractly. This collection is called *basis* since it generates some topology on X .

Definition 1.4. Topology generated by Basis

Let $\mathcal{B}$ is a basis of X . Then, define

$$\mathcal{T}_{\mathcal{B}} = \{U \subseteq X \mid \forall x \in U, \exists B \in \mathcal{B}, x \in B \subseteq U\}$$

Then $\mathcal{T}_{\mathcal{B}}$ is a topology on X and called topology generated by $\mathcal{B}$.

Remark 1.5

$\mathcal{T}_{\mathcal{B}}$ is well-defined topology.

As we noted $\mathcal{B}$ ‘generates’ some topology and it is called basis, there are more intuitive way to construct $\mathcal{T}_{\mathcal{B}}$.

Lemma 1.6. Topology generated by Basis

$\mathcal{T}_{\mathcal{B}}$ = collection of all unions in $\mathcal{B}$.

Then next the natural question is that, *can we compare 2 topologies with its basis?* The answer is yes, and we’ll use following theorem in many times. Following theorem allows us to compare the inclusion relation (or equivalence relation) of 2 topologies via its basis.

Theorem 1.7. Topology Comparison with Basis

Let $\mathcal{B}_1, \mathcal{B}_2$ be basis of X and $\mathcal{T}_1, \mathcal{T}_2$ be topology of X generated by $\mathcal{B}_1, \mathcal{B}_2$ respectively. Then $\mathcal{T}_1 \subseteq \mathcal{T}_2$ if and only if $\forall x \in X, \forall B_1 \in \mathcal{B}_1$ such that $x \in B_1$, there exists $B_2 \in \mathcal{B}_2$ such that $x \in B_2 \subseteq B_1$.

We observe that once we have basis $\mathcal{B}$, then we can generate some topology by unioning them. However, in 1.1 topology is closed under not only union but also finite intersection. So we can define more light basis for topology(or, basis for basis).

Definition 1.8. Subbasis

A subbasis $\mathcal{S}$ is collection of subsets of X having following axiom.

1. $\forall x \in X, \exists S \in \mathcal{S}$ such that $x \in S$.

equivalently, the entire union of $\mathcal{S}$ is X .

Then, we can observe that

$$\mathcal{S} \xrightarrow{\text{finite intersections}} \mathcal{B} \xrightarrow{\text{arbitrary unions}} \mathcal{T}$$

2 Some Basic Famous Topologies

In this section, we'll define a few useful topologies.

Definition 2.1. Standard Topology

Let $X = \mathbb{R}$ and $\mathcal{B} = \{(a, b) \mid a < b\}$. $\mathcal{T}_{\mathcal{B}}$ is called standard topology of $\mathbb{R}$.

Definition 2.2. Lower Limit Topology

Let $X = \mathbb{R}$ and $\mathcal{B} = \{[a, b) \mid a < b\}$. $\mathcal{T}_{\mathcal{B}}$ is called lower limit topology of $\mathbb{R}$.

Definition 2.3. Order Topology

Let X be a simply ordered set. i.e., $\forall x, y \in X$, $x < y$ or $x = y$ or $y < x$ holds and the relation $<$ is transitive. Then we define standard topology for X which is called order topology. Let

$$\mathcal{B} = \begin{cases} (a, b) & \forall a < b \in X \\ [m, b) & m \text{ is smallest element in } X \\ (a, M] & M \text{ is largest element in } X \end{cases}$$

Topology generated by this $\mathcal{B}$ on simply ordered set X is called *order topology*.

For order topology, we can define subbasis generates such basis. It is called open rays and define by

$$\mathcal{S} = \{(-\infty, a), (a, \infty) \mid a \in X\}$$

Remark 2.4

$\mathcal{S}$ is subbasis of $\mathcal{B}$.

One obvious step is defining 'sub' space and find its natural topology. The most natural definition is that, if Y is subspace of X , it inherits all open sets in X into Y . means that,

If Y is subspace of X , then open sets in X is also open in Y .

Definition 2.5. Subspace Topology

Let $(X, \mathcal{T}_X)$ is a topological space and $Y \subseteq X$. We define

$$\mathcal{T}_Y = \{U \cap Y \mid U \in \mathcal{T}_X\}$$

for topology of Y and call *subspace topology*. And $(Y, \mathcal{T}_Y)$ is called *subspace* of X .

Then similarly the basis of $\mathcal{T}_Y$ has same relation.

$$\mathcal{B}_Y = \{B \cap Y \mid B \in \mathcal{B}_X\} \text{ is basis for } \mathcal{T}_Y.$$

Warning 2.6

Open sets in general $Y \subseteq X$ do not need to also open in X .

Lemma 2.7. When open sets in Y is also open in X ?

If Y is subspace of X and Y is open in X , then open sets in Y is also open in X .

Proof. From definition, any open set $V \in \mathcal{T}_Y$ in Y is formed $V = U \cap Y$ and $U, Y \in \mathcal{T}_X$ so $V \in \mathcal{T}_X$. $\square$

Then also one natural question is arised. **How about subspace of simply ordered space?** Here interesting results are occur. We observe 3 subspace of $\mathbb{R}$.

$$Y_1 = [0, 1], \quad Y_2 = [0, 1) \cup \{2\}, \quad Y_3 = [0, 1] \cup [2, 3]$$

In Y_1, Y_3 , subspace topology of them are equivalent to ordered topology of themselves. However, in Y_2 , subspace topology is not equivalent to ordered topology since then singleton set $\{2\}$ is open in subspace topology but not in ordered topology. In general,

Theorem 2.8. Subspace Topology and Order Topology

Let $(X, \mathcal{T}_X)$ is ordered topology space and $Y \subseteq X$. $\mathcal{T}_1$ is subspace topology of Y and $\mathcal{T}_2$ is ordered topology of Y . (since Y also simply ordered) Then, $\mathcal{T}_2 \subseteq \mathcal{T}_1$ in general. Furthermore, $\mathcal{T}_1 = \mathcal{T}_2$ if and only if every gap in Y is either open or closed.

3 Closed Sets and Limit Points

We defined basic notions of topology via well-shaped subsets called *open sets*. Another type of sets can be useful.

Definition 3.1. Closed Sets

$C \subset X$ is closed when $X \setminus C$ is open in X

Openness and closedness is familiar notion in $\mathbb{R}^n$ space. Closed set is generalization of the closedness into general sets. Sometimes, arguing with closed set is much easier than open sets despite they are just complement of open sets.

Theorem 3.2. Topology Axioms for Closed Sets

Let $(X, \mathcal{T})$ be a topological space. Then following variation of axioms hold.

1. $\emptyset, X$ is closed set.
2. Arbitrary intersections of closed sets are closed.
3. Finite unions of closed sets are closed.

Theorem 3.3. Subspace Topology for Closed Sets

Let Y be subspace of X . Then, $D \subseteq Y$ is closed in Y if and only if there exists closed set C in X such that $D = C \cap Y$.

Proof. If $D \subseteq Y$ is closed, then $Y \setminus D$ is open in Y . So, we can find open set U in X such that $Y \setminus D = U \cap Y$. Then, $D = U^c \cap Y$ and U^c is closed in X . If $D = C \cap Y$ for closed set C in X , $Y \setminus D = Y \cap C^c$ is open in Y , so D is closed in Y . $\square$

Theorem 3.4. When closed sets in Y is also closed in X ?

If Y is subspace of X and Y is closed in X , then closed sets in Y is closed in X .

Now we define few more way to construct meaningful sets.

Definition 3.5. Interior and Closure of a Set

Let A be any subset of X . The interior of A (denoted by $\text{Int } A$) is union of all open sets contained in A . i.e., the largest open set in A . The closure of A (denoted by $\bar{A}$) is intersection of all closed sets containing A . i.e., the smallest closed set containing A .

In many times, closure of set acts quite important roles. So we state few theorems related to the closure of set.

Theorem 3.6. Closure in the Subspace

Let X be a topological space and Y is subspace of X . For $A \subseteq Y \subseteq X$, denote $\bar{A}$ by closure of A in X . Then closure of A in Y is determined by $\bar{A} \cap Y$.

Proof. Let C be closed set in X such that $A \subseteq C$. Then, $C \cap Y$ is closed in Y and $A \subseteq C \cap Y$. So, $\bar{A} \cap Y \subseteq \text{closure of } A \text{ in } Y$. And for D be closed set in Y such that $A \subseteq D$, by definition, $D = C \cap Y$ where C is closed and $A \subseteq C$. So, closure of A in Y is equivalent to $\bar{A} \cap Y$. $\square$

Theorem 3.7. Points in Closure

Let X be a topological space and $A \subseteq X$ be any subset. Then $x \in \bar{A}$ if and only if every *neighborhoods* of x *intersects* with A .

Here, *neighborhood* of x means that any open sets containing x . Also, we say A, B are intersects when $A \cap B \neq \emptyset$. This is important property of points in the closure.

Proof. If $x \in \bar{A}$ but $\exists U$ which is neighborhood of x and $U \cap A = \emptyset$, Then $A \subseteq U^c$ and U^c is closed. So by definition, $\bar{A} \subseteq U^c$ so $x \in U^c$ which is contradiction. If every neighborhoods of x intersects with A but $x \notin \bar{A}$, then it means that there exists a closed set C such that containing A but does not contain x . It implies that $x \in X \setminus C$ and $X \setminus C \cap A = \emptyset$ which is contradiction. $\square$

Definition 3.8. Limit Points

If A is a subset of topological space X , we say that x is a limit point of A when every neighborhoods of x intersects with A other than x itself. i.e., $x \in \overline{A \setminus \{x\}}$.

Then now, we can describe the closure of set A with A itself and its limit points.

Theorem 3.9. Closure Representation

Let X be a topological space and $A \subseteq X$ be any subset. Denote A' by set of limit points of A . Then,

$$\bar{A} = A \cup A'$$

In other words, $\bar{A} \setminus A = \text{limit points of } A \text{ but not in } A$.

Corollary 3.10. When A is closed?

$A \subseteq X$ is closed if and only if $A' \subseteq A$. i.e., A contains all of its limit points.

4 Basic notion of Hausdorff Space

We learned how can we define each shapes of sets and construct them into spaces. Then we can wonder, what is the good shapes for spaces? And how can we categorize them? Actually the answer for this question is dealt in further section but we define very part of them here since it will be heavily used.

Definition 4.1. Hausdorff Space

A topological space $(X, \mathcal{T})$ is called Hausdorff space when

$$\forall x \neq y \in X, \text{ there exists disjoint neighborhoods of } x, y \text{ respectively.}$$

In other words, we can *separate* two *points* via their neighborhoods always.

Actually we can understand that why this is good property for space since real line $\mathbb{R}$ have this property. Once our space is Hausdorff, we can find a few interesting facts.

Theorem 4.2. Finite sets in Hausdorff spaces

Let X be a Hausdorff space. Then, every finite set in X is closed.

Proof. Since finite union of closed sets are closed, so it is sufficient to show that any singleton set $\{x\}$ is closed in X . Now suppose that $y \in X$ is a limit point of $\{x\}$. If $x \neq y$, since X is Hausdorff we can find U_1, U_2 such that both are neighborhoods of x, y respectively and $U_1 \cap U_2 = \emptyset$. Then, according to theorem 3.7, y can't be limit point of $\{x\}$. So unless x is limit point or not, we can state $\{x\}$ contained all its limit points and so closed set. $\square$

Now we find next interesting property of Hausdorff space. Let's call the $\mathbb{N}$ -indexed sets of points in X by *sequences* and denoted by $\{x_n\}_{n \in \mathbb{N}}$. In general notion, we state that $\{x_n\}_{n \in \mathbb{N}}$ converges to $x \in X$ when

For any neighborhood U of x , we can find $N \in \mathbb{N}$ such that $x_m \in U$ for $m \geq N$.

Then following good property holds in Hausdorff space.

Theorem 4.3. Unique Convergency in Hausdorff Space

If X is a Hausdorff space, then any sequences in X converges into at most one point.

Theorem 4.4. Hausdorff spaces with Famous Topologies

Any space with Order topology, Subspace topology of Hausdorff space is Hausdorff.

5 Continuous Functions and Homeomorphisms

From highschool to undergrad analysis courses, the continuity of function is matter. By the way, the concept of continuity in $\mathbb{R}^n$ (in general, metric space) is quite obvious. Our goal is generalize them into general topological space. One beautiful definition is following.

Definition 5.1. Continuous Function

Let X, Y be any topological spaces. A function $f : X \rightarrow Y$ is *continuous* when for any open set V in Y , the preimage $f^{-1}(V) \subseteq X$ is also open set in X .

Remark 5.2

Continuity is depends on not only f itself but also topologies of X and Y .

Here, I mention why we take preimage for define continuity. First at all, this notion is called **pull-back**. This is important since it preserves many structural things in codomain space Y .

Proposition 5.3. Preimage Propositions

1. $f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B)$
2. $f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B)$
3. $f^{-1}(A^c) = (f^{-1}(A))^c$

These are not true in original mapping f . So, we can analyze topological structures of Y in X via pull-back. Also, the natural notion ε - δ definition of continuity also refer the preimage. Once we view this definition as just generalization of ε - δ definition, it seemed be natural.

Lemma 5.4. Checking Continuity

Let $\mathcal{S}$ be subbasis of topological space Y . Then for any $S \in \mathcal{S}$, if $f^{-1}(S)$ is open set in X , f is continuous.

Proof. Every open sets in Y can be generated by arbitrary union and finite intersection of elements in $\mathcal{S}$. Since 5.3 holds for preimages, if every $f^{-1}(S)$ is open then every preimage of open set is also open in X . $\square$

Theorem 5.5. Equivalence statements for Continuity

The followings are equivalent.

1. f is continuous.
2. For any $A \subset X$, $f(\bar{A}) \subseteq \overline{f(A)}$
3. For every closed set C of Y , $f^{-1}(C)$ is closed in X .
4. For each $x \in X$, each neighborhood V of $f(x)$, there exists neighborhood U of x such that $f(U) \subseteq V$.

(4) is just changing of forms of statement into ε - δ definition form. The remarkable one is second one. It can be written: If x is a limit point of A in X , then $f(x)$ is a limit point of $f(A)$ in Y or equivalently, continuous map preserves the limit point relation.

Definition 5.6. Homeomorphism

Let X, Y be topological spaces and $f : X \rightarrow Y$ is a bijection. If both f, f^{-1} are continuous then f is called *homeomorphism* and X, Y are *homeomorphic*.

The homeomorphism f between X and Y construct bijection between $\mathcal{T}_X$ and $\mathcal{T}_Y$ also. i.e., open sets in both spaces are one-to-one corresponds. We say that some property is topological property if it preserves by homeomorphisms. And also we say that an injective continuous map $f : X \rightarrow Y$ is *embedding*.

Theorem 5.7. Constructing Continuous Functions

Let X, Y and Z be topological spaces.

1. Any constant function is continuous.
2. If A is a subspace of X , the natural inclusion $j : A \hookrightarrow X$ is continuous.
3. If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are continuous, then $h = g \circ f$ is continuous.
4. If $f : X \rightarrow Y$ is continuous then for subspace A of X , $f|_A : A \rightarrow Y$ is continuous.
5. Let $f : X \rightarrow Y$ is continuous. If $\text{Im}(X)$ is subspace of Y , then $g : X \rightarrow \text{Im}(X)$ is continuous. If Y is subspace of Z , then $h : X \rightarrow Z$ is continuous.
6. If $X = \bigcup U_\alpha$ and every $f|_{U_\alpha}$ is continuous, then $f : X \rightarrow Y$ is continuous.

These are not hard to follow and prove. We state important lemma here.

Lemma 5.8. Pasting Lemma

Let X, Y be topological spaces and $X = A \cup B$ where A, B are closed subspaces in X . Let $f : A \rightarrow Y$ and $g : B \rightarrow Y$ be continuous. If $f(x) = g(x)$ for every $x \in A \cap B$, then we can define continuous function $h : X \rightarrow Y$ defined by

$$h(x) = \begin{cases} f(x) & x \in A \\ g(x) & x \in B \end{cases}$$

This is well-defined since we assume $f(x) = g(x)$ for $x \in A \cap B$.

Proof. As we stated, $h : X \rightarrow Y$ is well-defined. Let C be any closed set in Y . Then, $f^{-1}(C), g^{-1}(C)$ are closed in A, B respectively by 5.5. However since A, B are closed subspace of X , $f^{-1}(C), g^{-1}(C)$ is closed in X by 3.4. However,

$$h^{-1}(C) = f^{-1}(C) \cup g^{-1}(C) \implies \text{closed in } X$$

So, $h : X \rightarrow Y$ is continuous function. □

6 The Product Topology

We often consider Cartesian product of sets(or spaces). It is good time to state how Cartesian product can be generalized into topological concepts. We'll see familiar product $X \times Y$ first and then generalize into arbitrary Cartesian product $\prod_{\alpha \in J} X_\alpha$ for index set J .

Definition 6.1. Simple Cartesian Product Space $X \times Y$

Let $(X, \mathcal{T}_X), (Y, \mathcal{T}_Y)$ be a topological spaces. For the Cartesian product $X \times Y$, we define *product topology* generated by following basis.

$$\mathcal{B} := \{U \times V \mid U \in \mathcal{T}_X, V \in \mathcal{T}_Y\}$$

Remark 6.2

We take $\mathcal{T}_X \times \mathcal{T}_Y$ as just a **basis** for $X \times Y$. So, the *product topology* on $X \times Y$ is arbitrary unions of $\mathcal{T}_X \times \mathcal{T}_Y$. Intuitively, imagine the $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$ space. The 'shapes' in $\mathbb{R}^2$ is more various than just $\mathcal{T}_{\mathbb{R}} \times \mathcal{T}_{\mathbb{R}}$. So, $\mathcal{T}_X \times \mathcal{T}_Y$ just basis generates product topology.

Here we can check a few properties that we stated in previous sections.

Lemma 6.3. Basis of Product Topology

Let X, Y be topological spaces. If $\mathcal{B}_X, \mathcal{B}_Y$ are basis for X, Y respectively, then $\mathcal{B} = \mathcal{B}_X \times \mathcal{B}_Y$ is a basis for product topology of $X \times Y$.

Definition 6.4. Projection

We define $\pi_1 : X \times Y \rightarrow X$ and $\pi_2 : X \times Y \rightarrow Y$ in natural manner.

Lemma 6.5. Subbasis of Product Topology

Let X, Y be topological spaces. Following is subbasis for product topology of $X \times Y$:

$$\mathcal{S} = \{\pi_1^{-1}(U) \mid U \text{ is open in } X\} \cup \{\pi_2^{-1}(V) \mid V \text{ is open in } Y\}$$

Lemma 6.6. Subspace of Product Space

Let X, Y be topological spaces. If A, B are subspaces of X, Y respectively, then $A \times B$ is subspace of $X \times Y$.

Lemma 6.7. Product of Hausdorff is Hausdorff

If X, Y are Hausdorff space, then $X \times Y$ with product topology is Hausdorff.

These are not hard to understand. We can easily generalize this into finite Cartesian product by

$$X_1 \times X_2 \times \cdots \times X_n$$

Then here 6.5 still holds. One natural following step is considering the arbitrary Cartesian products.

Definition 6.8. Indexing, J -tuple and Coordinates

Let J be an arbitrary index set. Given a set X , we define a J -tuple of elements of X to be a function

$$\mathbf{x} : J \rightarrow X$$

We often denote $\mathbf{x}(\alpha)$ by x_α and call it the α -th coordinate of $\mathbf{x}$.

With this arbitrary index set J , we can define general Cartesian product.

Definition 6.9. Cartesian Product

Let J be an arbitrary index set and $\{A_\alpha\}_{\alpha \in J}$ be an indexed family of sets. The Cartesian product of this indexed family is denoted by

$$\prod_{\alpha \in J} A_\alpha = \{\text{All } J\text{-tuples } \{x_\alpha\}_{\alpha \in J} \text{ where each } x_\alpha \in A_\alpha\}$$

Now we can define this notion into topological Cartesian product space. However, there are 2 ways to construct topology on such space. Actually, one is quite natural but less useful, and the other is related to 6.5. It means that 6.5 is not the case in general, and two topologies are distinct when J is an infinite index set.

Definition 6.10. Box Topology

Given arbitrary index set J and $\{X_\alpha\}_{\alpha \in J}$ be a family of topological spaces. Then, we can find Cartesian product $\prod_{\alpha \in J} X_\alpha$. Then we can define a basis in following way:

$$\mathcal{B} = \left\{ \prod_{\alpha \in J} U_\alpha \mid U_\alpha \text{ is open in } X_\alpha \right\}$$

Then this is topology on $\prod_{\alpha \in J} X_\alpha$ and called *box topology*.

Definition 6.11. Projection

For each $\beta \in J$, we define

$$\pi_\beta : \prod_{\alpha \in J} X_\alpha \rightarrow X_\beta$$

defined by

$$\pi_\beta : \mathbf{x} \mapsto \mathbf{x}(\beta)$$

This is called the projection mapping associated with the index β .

We generalize the notion of projection in 6.4 into general Cartesian product.

Definition 6.12. Product Topology

We can define a subbasis $\mathcal{S}$ of $\prod_{\alpha \in J} X_\alpha$ in following way:

$$\mathcal{S}_\beta = \{\pi_\beta^{-1}(U_\beta) \mid U_\beta \text{ is open in } X_\beta\} \implies \mathcal{S} = \bigcup_{\beta \in J} \mathcal{S}_\beta$$

The topology generated by this subbasis is called the *product topology*. And with this, $\prod_{\alpha \in J} X_\alpha$ is called *product space*.

Compare these two topology is not quite difficult. As we stated in 1.7, it is enough to show the differences between the basis of box topology and the basis of product topology.

Proposition 6.13. Comparison between Box topology and Product Topology

The basis of box topology contains all possible forms of $\prod_{\alpha \in J} U_\alpha$ where U_α is any open set in X_α . However, the basis of product topology is obtained by **finite intersections** of subbasis. So, the basis of product topology contains only has the following form of elements:

$$\pi_{\beta_1}^{-1}(U_{\beta_1}) \cap \cdots \cap \pi_{\beta_n}^{-1}(U_{\beta_n}) = \prod_{\alpha \in J} U_\alpha, \quad U_\alpha = \begin{cases} X_\alpha & \alpha \notin \{\beta_1, \dots, \beta_n\} \\ U_{\beta_i} & i \in [n] \end{cases}$$

So the basis of box topology contains the basis of product topology, so *box topology is finer than product topology*.

According to the textbook [2], there is an answer **why we generally use the definition of product topology**. i.e., despite the definition of box topology is more finer and stronger, we assume that the topology of $\prod_{\alpha \in J} X_\alpha$ will be product topology in general. It is because, there are many important theorems in topology such that holds for product topology, but fails for box topology. **However note that when index set J is infinite then matter. For finite Cartesian product of spaces, there are not any confusion.**

Now let's state bunch of theorems here.

Theorem 6.14. Basis generated by Basis of each Space

Suppose that $\{X_\alpha\}_{\alpha \in J}$ be family of spaces and $\mathcal{B}_\alpha$ is the basis of X_α . Then, the basis of *box topology* is given by

$$\mathcal{B}_{\text{box}} = \left\{ \prod_{\alpha \in J} B_\alpha \mid B_\alpha \in \mathcal{B}_\alpha \right\}$$

The basis of *product topology* is given by

$$\mathcal{B}_{\text{prod}} = \left\{ \prod_{\alpha \in J} B_\alpha \mid B_\alpha = \begin{cases} B_\alpha \in \mathcal{B}_\alpha & \text{Finitely many } \alpha \\ X_\alpha & \text{otherwise} \end{cases} \right\}$$

Theorem 6.15. Subspace Topology and Box and Product Topology

Let A_α be a subspace of X_α for each $\alpha \in J$. Now box topology of $\prod_{\alpha \in J} A_\alpha$ is subspace of box topology of $\prod_{\alpha \in J} X_\alpha$. And same for product topology.

Theorem 6.16. Hausdorff in Box and Product Topology

If each X_α is Hausdorff, $\prod_{\alpha \in J} X_\alpha$ is also Hausdorff in both box and product topologies.

Theorem 6.17. Closure in Box and Product Topology

Let $A_\alpha \subseteq X_\alpha$ for each $\alpha \in J$. If $\prod X_\alpha$ is given either the product or the box topology, then

$$\prod \overline{A_\alpha} = \overline{\prod A_\alpha}$$

Proof. Suppose that $\mathbf{x} = (x_\alpha)_{\alpha \in J} \in \prod \overline{A_\alpha}$. Then, let $\prod U_\alpha$ be any basis neighborhood of $\mathbf{x}$ in box or product topology of $\prod X_\alpha$. Then each U_α is neighborhood of x_α in X_α , and since $x_\alpha \in \overline{A_\alpha}$ so we can find $y_\alpha \in A_\alpha \cap U_\alpha$. So, $\mathbf{y} = (y_\alpha)_{\alpha \in J} \in \prod U_\alpha \cap \prod A_\alpha$. So $\mathbf{x} \in \overline{\prod A_\alpha}$.

Suppose that $\mathbf{x} \in \overline{\prod A_\alpha}$. Let V_β be an arbitrary open set of X_β containing x_β . Since $\pi_\beta^{-1}(V_\beta)$ is open in $\prod A_\alpha$ in both topology and containing $\mathbf{x}$, there exists $\mathbf{y} \in \pi_\beta^{-1}(V_\beta) \cap \prod A_\alpha$. This implies that, $y_\beta \in V_\beta \cap A_\beta$. So, $x_\beta \in \overline{A_\beta}$. $\square$

Following is the first situation when we can observe box topology and product topology is different.

Theorem 6.18. Continuity in Product Topology

Let $f : A \rightarrow \prod X_\alpha$ be given by the equation

$$f(a) = (f_\alpha(a))_{\alpha \in J}$$

where $f_\alpha : A \rightarrow X_\alpha$ for each α . Let $\prod X_\alpha$ have the product topology. Then the function f is continuous if and only if each function f_α is continuous.

Proof. ($\Rightarrow$) Suppose that f is continuous. Then for any open set U_β in A_β , we can define following open set in $\prod X_\alpha$,

$$U^* = \prod U_\alpha = \begin{cases} U_\beta & \alpha = \beta \\ X_\alpha & \alpha \neq \beta \end{cases}$$

Then this is a basis element of product topology of $\prod X_\alpha$, hence open. Since f is continuous mapping, $f^{-1}(U^*) = f_\beta^{-1}(U_\beta)$ is open in A . So each f_β is continuous.

($\Leftarrow$) Suppose that each f_α is continuous. According to 5.4 it is sufficient to show that preimage of subbasis is open in A . Each subbasis has the form

$$\pi_\beta^{-1}(V_\beta), \quad V_\beta \text{ is open in } X_\beta$$

So that, the preimage of each subbasis element has the form

$$f^{-1}(\pi_\beta^{-1}(V_\beta)) = f_\beta^{-1}(V_\beta) \implies \text{open in } A \text{ since } f_\beta \text{ is continuous}$$

Thus, f is continuous mapping. □

7 The Quotient Topology

The notion of quotient is always non-trivial (But, very fun if once you understand). In the Group theory, we encounter the quotient group with normal subgroups. The intuitive concept was that, grouping some subset into sort of equivalent classes. We can do same thing in topology. Whenever the space is X , we can grouping some elements in equivalent classes. Such notion is visualizable and natural, see following figure.

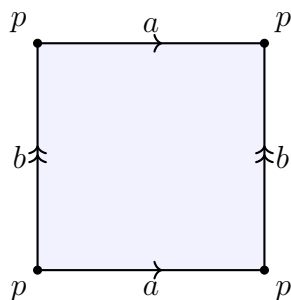


Figure 1: The fundamental polygon of a torus.

Now suppose that 2 sides represented as b are equivalent. Then the natural image of such rectangle can be represented as following.

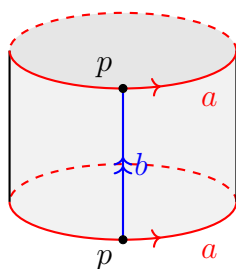


Figure 2: The cylinder formed by identifying the edges b of the fundamental polygon.

Now, do same thing to the both side a also.

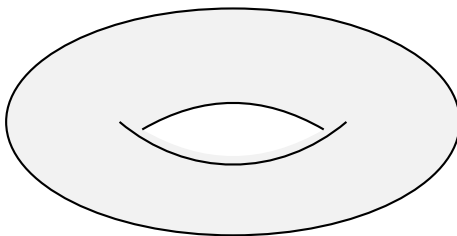


Figure 3: The standard generators a and b on the torus $\mathbb{T}^2$.

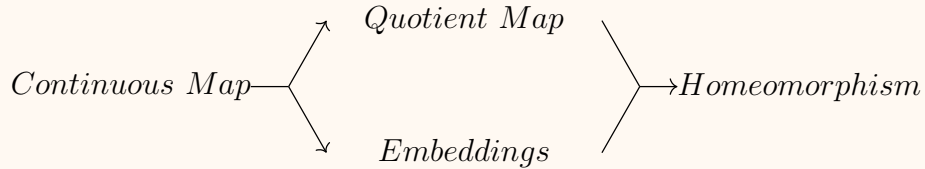
Our goal in this section is generalizing this notions into topological words. Let's define and understand bunch of things from now on.

Definition 7.1. Quotient Map

Let X, Y be topological spaces and $p : X \rightarrow Y$ be a surjective map. The map p is said to be a *quotient map* provided a subset U of Y is open in Y if and only if $p^{-1}(U)$ is open in X .

Remark 7.2

Following is the order of strongness of definitions.



Quotient map is continuous map with surjectivity and Embedding is continuous map with injectivity, and Homeomorphism is continuous map with bijectivity.

The if and only if condition above implies that, quotient map p is continuous($\Rightarrow$) and if preimage $p^{-1}(U)$ of something $U \subseteq Y$ is open in X , then U is open in Y . ($\Leftarrow$) **Note** that, there aren't any guarantee that image of any open set in X is also open in Y . We can obtain an alternative way to define quotient maps.

Definition 7.3. Saturated Set

Let X, Y be topological spaces and $p : X \rightarrow Y$ is a surjective map. We say that $C \subseteq X$ is saturated with respect to p if C equals the complete preimage of some subset of Y . i.e., $p^{-1}(p(C)) = C$ holds. Equivalently, C contains all $p^{-1}(\{y\})$ such that intersects with C .

Then now we can say that p is quotient map if and only if p is continuous and every open saturated set in X map to open set in Y . We also say that $f : X \rightarrow Y$ is an *open map* if image of open set in X is open in Y . Then, every surjective continuous open map is quotient map. (First one is stronger notion). Before move on, we summarize such confusing concepts in following way.

Remark 7.4. Continuous, Quotient, Homeomorphism

- If f is continuous, $U \subseteq Y$ is open $\implies$ complete preimage of U is open in X .
- If f is quotient map, there is a bijection between open sets in Y and open complete preimages in X .
- If f is Homeomorphism, there is a bijection between open sets in Y and open sets in X .

Now we can define *quotient topology* and *quotient space* which are main topics in this section.

Definition 7.5. Quotient Topology

X is a space and A is a set. For any $p : X \rightarrow A$ surjective map, there exists unique topology of A such that make p be a quotient map. Such topology is called *quotient topology* induced by p .

Such topology of A is just collection of $U \subseteq A$ such that $p^{-1}(U)$ is open in X . It automatically makes that p be a continuous map, and even quotient map. Intuitively, surjective map can be imagine as a compression of X into smaller set A . Above definition states that, there is the unique ‘shape’ for A that makes this becomes ‘good’ compression. Actually the ‘compression’ of space X is quite non-trivial in above context. We define the *quotient space* in general and see how to describe such compression in topological words.

Definition 7.6. Quotient Space

Let X be a topological space and X^* be a disjoint partition of X . We can define the projection $p : X \rightarrow X^*$ in natural manner, which is surjective map. Then 7.5 says that we can find unique *quotient topology* of X^* . With it, X^* is called quotient space of X .

Then it is not hard to understand that X^* is compressed space of space X . Each element in X^* is a sort of *representation* of its equivalent classes of elements of X . So any $U \subseteq X^*$ is collection of *equivalent classes* and its preimage $p^{-1}(U)$ is open.

As we stated in first of this section, now we can generalize the constructing torus from 2D rectangle. Let

$$X = [0, 1] \times [0, 1]$$

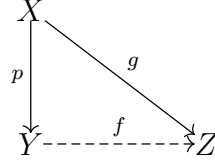
Then we can define X^* by:

$$X^* = \begin{cases} \{(x, y)\} & (x, y) \in (0, 1) \times (0, 1) \\ \{(0, y), (1, y)\} & y \in (0, 1) \\ \{(x, 0), (x, 1)\} & x \in (0, 1) \\ \{(0, 0), (0, 1), (1, 0), (1, 1)\} \end{cases}$$

Then X^* becomes a torus and we can pull-back open sets on torus into open sets on rectangle. We state 2 theorems related to continuity on quotient maps.

Theorem 7.7. Continuity of Quotient Map - Mapping Out

Let $p : X \rightarrow Y$ be a quotient map. Let Z be a space and let $g : X \rightarrow Z$ be a map such that constant on each $p^{-1}(y)$ for $y \in Y$. Then, g induces a map $f : Y \rightarrow Z$ such that $f \circ p = g$. f is continuous if and only if g is continuous, f is quotient map if and only if g is quotient map.



Proof. f can be well-defined as following way:

$$f : y \mapsto g(p^{-1}(y))$$

Then for any $x \in X$,

$$f \circ p(x) = g(p^{-1}p(x)) = g(x)$$

Then it is clear that if f is continuous(quotient map), then g is continuous(quotient map). Now suppose that g is continuous. Let U be any open set in Z . Then, $g^{-1}(U)$ is open in X . However, $g^{-1}(U)$ is complete preimage of some subset V in Y so $p^{-1}(V) = g^{-1}(U)$. Since p is quotient map, V is open in Y . However, $V = p \circ g^{-1}(U) = f^{-1}(U)$ is open in Y , so f is continuous. $\square$

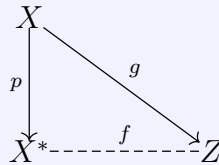
Corollary 7.8

Let $g : X \rightarrow Z$ be a surjective continuous map. Let X^* be the following collection of subsets of X :

$$X^* = \{g^{-1}(z) \mid z \in Z\}$$

Give X^* the quotient topology.

- The map g induces a bijective continuous map $f : X^* \rightarrow Z$, which is a homeomorphism if and only if g is a quotient map.



- If Z is Hausdorff, so is X^* .

Proof. Since g is surjective, $f = g \circ p^{-1}$ clearly forms the bijection between X^* and Z . Suppose that f is a homeomorphism and $g^{-1}(U)$ is open for some subset U of Z . Since f is bijection, there exists V in X^* such that $f(V) = U$. Then by definition of

f , $p^{-1}(V) = g^{-1}(U)$. So $p^{-1}(V)$ is open in X , implies that V is open in X^* . Since f is homeomorphism, U is open in Z . Suppose that g is a quotient map. Then for any open set U in X , $p^{-1}(U)$ and $g^{-1}(U)$ both are open in X^*, Z respectively. However, they bijects via f , so f is homeomorphism. If Z is Hausdorff, X^* is clearly Hausdorff. $\square$

8 Connected Spaces

From now on, our main focus is on defining and analyzing ‘good’ shapes and spaces in general. First of all, we can imagine the good properties in $\mathbb{R}^n$ for example, *Maximum Value Theorem*, *Intermediate Value Theorem*, *Uniform Continuity Theorem* so on. However, in $\mathbb{R}$, they holds for closed interval $[a, b]$. So we can imagine that $[a, b]$ is somewhat good-shaped subspace of the good space $\mathbb{R}$. Since the condition for ‘good’ can be differ for each properties, so we want to generalize them as much as possible. The first concept is quite natural, which is called *connectedness*.

Definition 8.1. Seperation and Connected

Let X be a topological space. We say that U, V is a *seperation* of X if $U, V \neq \emptyset$, $X = U \cup V$ and U, V are open set in X . If such U, V are not exists in X , then we say that X is *connected* space.

Connectedness is obviously topological invariant. i.e., any Homeomorphism preserves the connectivity. This is clear since any Homeomorphisms preserves the seperation. We can define connected space slightly differently.

Lemma 8.2

X is connected if and only if clopen sets in X are only the trivial ones(i.e., if $A \subseteq X$ is both closed and open, then $A = \emptyset$ or X).

Proof. Existance of clopen set is equivalent to existance of seperation of X . $\square$

Lemma 8.3. Subspace Connectivity

If Y is a subspace of X , a seperation of Y is a pair of disjoint nonempty sets A and B whose union is Y , neither of which contains a limit point of the other. The space Y is connected if and only if there exists no seperation of Y .

Proof. Suppose that Y is not connected, then there exists $A, B \neq \emptyset$, $A \cup B = Y$ and A, B are open in Y . Then, A, B are also closed in Y . So, by 3.6,

$$A = \text{closure of } A \text{ in } Y = \bar{A} \cap Y$$

And similarly for B , we can obtain that A, B contains all of its limit points in themselves. So, A, B is the seperation of Y . Suppose that A, B are a seperation of Y . Then we can obtain A, B are both closed and open in Y , so Y is not connected. $\square$

Here, we can imagine that not only elements but also a limit point acts role of *connection port* of a set. First, we can state the notion of *connection port* of elements as following lemmata.

Lemma 8.4. Connected Subspaces

If the sets C and D form a separation of X , and if Y is a connected subspace of X , then Y lies entirely within either C or D .

Proof. Suppose not, then $C \cap Y, D \cap Y \neq \emptyset$ and $(C \cap Y) \cap (D \cap Y) = \emptyset$ and both are open in Y , moreover $(C \cap Y) \cup (D \cap Y) = Y$. So, contradiction to Y is connected subspace. $\square$

Lemma 8.5. Connection Port - Elements

The union of connected subspaces that **point in common** is connected.

Proof. Let $\{A_\alpha\}_{\alpha \in J}$ be a family of connected subspaces of X and there exists $x \in X$ such that $x \in A_\alpha$ for any $\alpha \in J$. Suppose that $\bigcup A_\alpha$ is not connected. Then we can find $\bigcup A_\alpha = C \cup D$ where C, D forms separation of $\bigcup A_\alpha$. W.L.O.G., $x \in C$. Then, according to ??, every $A_\alpha \subseteq C$ so $D = \emptyset$ which is contradiction. $\square$

Now we state the notion of connection port of limit points as following theorem.

Theorem 8.6. Connection Port - Limit Points

Let A be connected subspace of X . For $A \subseteq B \subseteq \bar{A}$, then B is connected subspace.

Proof. Suppose that B is not connected. Then, we can find separation U, V of B such that U contains all its limit points in B and V also. However since $A \subseteq B$, according to 8.4 A entirely in U without loss of generality. Then all elements in V is $B \setminus A \subseteq A'$. But they are limit points of $A \subseteq U$, contradiction. $\square$

Now we'll see important results for connected spaces. The big question is, **what preserves the connectedness?**

Theorem 8.7. Continuous Map Preserves Connectedness

The image of connected set under continuous map is connected.

Proof. Let $f : X \rightarrow Y$ is continuous map and X is connected space. For convenience, let $Y = \text{Im}(X)$. If Y is not connected, there are U, V forming separation of Y . Since they are open, $f^{-1}(U), f^{-1}(V)$ also open in X . And they are non-empty and disjoint partition of X , contradiction to X is connected. $\square$

Theorem 8.8. Finite Cartesian Product of connected spaces

A finite Cartesian product of connected spaces is connected.

Proof. It is enough to show that if X, Y are connected spaces, then $X \times Y$ is connected. Pick one point $a \times b$ in $X \times Y$. For any $x \in X$, since $x \times Y \cong Y$ and $X \times b \cong X$, both

of $x \times Y, X \times b$ are connected. Then 8.5 implies that following subspace is connected:

$$T_x = (X \times b) \cup (x \times Y)$$

Now, form the union $\bigcup_{x \in X} T_x$ then this is also connected since $a \times b \in T_x$ for any $x \in X$ and then 8.5. However, $\bigcup_{x \in X} T_x = X \times Y$. So $X \times Y$ is connected. $\square$

Here are some interesting results. As we referred in 6.13, when we consider the arbitrary cartesian products, the connectedness depends on what topology we take for the space.

Theorem 8.9. Box Topology and Product Topology, in connectedness

Let $\{X_\alpha\}_{\alpha \in J}$ is family of connected spaces. $\prod_{\alpha \in J} X_\alpha$ is connected under product topology, but not always true under box topology.

The example is quite tricky. We can imagine $\mathbb{R}^\omega$ as a set of sequences on $\mathbb{R}$. Then, we can categorize such sequences by:

A : Bounded sequences , B : Unbounded sequences

Then $A \cap B = \emptyset$, $A \cup B = \mathbb{R}^\omega$. Now we claim that A, B both are open. For sequence $\{x_n\}_{n \in \mathbb{N}}$, construct

$$U = (x_1 - 1, x_1 + 1) \times (x_2 - 1, x_2 + 1) \times \cdots$$

Then U is open in box topology, and $U \subseteq A$ if $\{x_n\}$ is bounded, and $U \subseteq B$ if $\{x_n\}$ is unbounded. So A, B both are open and $\mathbb{R}^\omega$ is not connected. Actually this example shows that second part of 8.9.

Proof. Choose $(x_\alpha)_{\alpha \in J}$ in $\prod X_\alpha$. We can define Y by set of $(y_\alpha)_{\alpha \in J}$ such that $y_\alpha \neq x_\alpha$ for finitely many $\alpha \in J$. Then, we can define X_K by $\prod H_\alpha$ such that $H_\alpha = X_\alpha$ if $\alpha \in K$ and $H_\alpha = \{x_\alpha\}$ if not. Then $X_K \cong \prod_{\alpha \in K} X_\alpha$ Here if K is finite, then $\prod_{\alpha \in K} X_\alpha$ is connected by 8.8 and so for X_K . Note that $Y = \bigcup_{K \subseteq J, |K| < \infty} X_K$ and each X_K has common point (x_α) . So Y is connected. Now suppose any basis element $\prod_{\alpha \in L, |L| < \infty} U_\alpha$ for $\prod X_\alpha$. In such basis element, we can always find

$$(z_\alpha) = \begin{cases} x_\alpha & \alpha \notin L \\ z_\alpha & \alpha \in L \end{cases}$$

Then it is clear that $(z_\alpha) \in \prod_{\alpha \in L, |L| < \infty} U_\alpha$ and $(z_\alpha) \in Y$. So, every basis of $\prod X_\alpha$ intersects Y . Means that every points of $\prod X_\alpha$ is limit points of Y , implies that $\bar{Y} = \prod X_\alpha$. However Y is connected, so $\bar{Y}$ also connected by 8.6. So $\prod X_\alpha$ also connected. $\square$

9 Connected Subspaces of The Real Line

In this section, we state full-of intuitive parts about connectedness. We intuitively knows that any intervals and rays are connected in $\mathbb{R}$. Such connected sets in $\mathbb{R}$ has ‘good’ properties, like *intermediate value theorem*. Our goal is find the general condition that makes our space holds *intermediate value theorem*, like $\mathbb{R}$. And also, state another fully intuitive notion called *path connected*.

Definition 9.1. Linear continuum

A simply ordered set L non-empty is called a *linear continuum* if following holds:

1. L has the least upper bound property.
2. If $x < y$, exists z such that $x < z < y$.

Where least upper bound property means that, if $A \subseteq L$ is bounded above, then $\sup A \in L$.

This is the target property that our general space behaviors like $\mathbb{R}$ (in connectedness’ perspective). First $\mathbb{R}$ -like behavior is following.

Theorem 9.2. Convex Connectedness

Let L is a linear continuum with ordered topology. Any convex set Y in L is connected. So, L itself, intervals, and rays are connected.

Proof. Suppose that our convex set $Y \subseteq L$ is not connected. Then Y has a separation A, B . Let’s choose $a \in A, b \in B$ and $a < b$ without loss of generality. Since Y is convex, $[a, b] \subseteq Y$. Let’s denote

$$A_0 = [a, b] \cap A, \quad B_0 = [a, b] \cap B$$

However, $[a, b] \subseteq Y \subseteq L$ also supspace and A_0, B_0 are non-empty open sets in $[a, b]$, which union is $[a, b]$. So A_0, B_0 forms separation of $[a, b]$. Now, let $x = \sup A_0$. This is well-defined since A_0 is bounded and L is linear continuum. And $a \leq x \leq b$, so $x \in [a, b]$. Suppose that $x \in B_0$. Since B_0 is open, we can find small open interval for x in B_0 . See it partly, we obtain $(d, x] \subseteq B_0$. Then d is smaller upperbound than x , contradiction. Vice versa, Y is connected. $\square$

Now it’s time to generalize one of the famous theorem in real analysis, which is *intermediate value theorem*. The condition is much generalized, f is continuous, X is connected, Y is order space.

Theorem 9.3. The Intermediate Value Theorem

Let $f : X \rightarrow Y$ be a continuous map where X is connected and Y is order space. For $a, b \in X$, if $f(a) < r < f(b)$ then there exists $c \in X$ such that $f(c) = r$.

Proof. Suppose that such c is not exists. Then, $r \notin \text{Im}(X)$ so we can separate $\text{Im}(X)$ by

$$A_0 = \text{Im}(X) \cap (-\infty, r), \quad B_0 = \text{Im}(X) \cap (r, \infty)$$

Since A_0, B_0 is open in $\text{Im}(X)$ and non-empty disjoint, they are separation of $\text{Im}(X)$. However, the image of connected space under continuous map is connected by 8.7. So this is contradiction, and such c is exists. $\square$

Now we define another version of connectedness, which is called *path connected*. This is drawble concepts, but not exactly equivalent to what we defined.

Definition 9.4. Path Connected

Let X be a topological space. We define a *path* between $x, y \in X$ by a continuous mapping $f : [a, b] \rightarrow X$ such that $f(a) = x, f(b) = y$. Where $[a, b] \subseteq \mathbb{R}$. X is called *path connected* if for every $x, y \in X$, there exists a path from x to y .

Theorem 9.5. Path Connected Implies Connected

If X is path connected space, then X is connected space.

Proof. Suppose that X is not connected so there is a separation A, B of X . Then since every path is image of convex(so, connected) space, *path is connected*. So by 8.4, path is entirely in A or B . Then, we can't find a path between $a \in A$ and $b \in B$, contradiction to X is path connected. So, X is connected. $\square$

Proposition 9.6. Connected Does Not Implies Path Connected

Let $S = \{x \cdot \sin(1/x) \mid 0 < x \leq 1\}$. $\bar{S}$ is connected but not path connected.

Proof. Since S is image of $(0, 1]$ under continuous map, so S is connected. This implies that $\bar{S}$ is connected by 8.6. Note that $\bar{S} = S \cup (0 \times [-1, 1])$. There is not any path between $a \in S, b \in 0 \times [-1, 1]$. If there exists a path, $f : [0, 1] \rightarrow \bar{S}$ such that $f(0) = b$ and $f(1) = a$ is continuous. However, it oscillate in $[-1, 1]$ near $x = 0$, continuity is not hold near $x = 0$. $\square$

10 Compact Space

When we say something is ‘compact’, it means that ‘Wow, that’s very good or suitable’. According to our textbook page 161, Munkres[2] refer good historical development of the word ‘compact’ in topology. First, in $\mathbb{R}$, we observe that any closed interval has very good properties like uniform continuity theorem and maximum value theorem. After that, mathematicians like Weierstrass observe that every infinite subsets of $[a, b]$ has a limit point. Later, finally we find the core generalization via open coverings. We begin with define open covering of space.

Definition 10.1. Open Covering

A family $\mathcal{A}$ of subsets of $(X, \mathcal{T})$ is called *open covering* of X if every $A \in \mathcal{A}$ is open and $\bigcup_{A \in \mathcal{A}} A = X$.

With this definition, we define general *compact* in topological spaces.

Definition 10.2. Compact

X is said to be *compact* if for any open covering $\mathcal{A}$ of X contains *finite subcollection* that also covers X .

You might realize that it is hard to specify given space is compact. To decide that given space is *not* compact, we tries to find proper open covering that we never find finite subcovering from there. Often choose *sequential* covering such that decreases, so if there exists finite subcovering, some contradiction occurs. For example, following open covering for $(0, 1]$ with standard topology shows this:

$$\{(1/n, 1] \mid n \in \mathbb{N}\} \implies (0, 1] \text{ is not compact}$$

We can state some lemmata between previous concepts and compactness.

Lemma 10.3. Compactness under Union and Intersection

Finite union of compact subspaces are compact. Arbitrary intersections of compact subspaces in *Hausdorff* space is compact.

Lemma 10.4. Subspace Compactness

Let Y be a subspace of X . Y is compact if and only if every open covering of Y in X contains a finite subcovering of Y .

Theorem 10.5. Closed in Compact is Compact

Closed subspace of compact space is compact.

Theorem 10.6. Compact in Hausdorff is Closed

Compact subspace of Hausdorff space is closed.

Proof. Let X be a Hausdorff space and C is subspace of X which is compact. Then, we'll show that $X \setminus C$ is open in X . Choose any $x \in X \setminus C$. Then for any $y \in C$, since X is Hausdorff, we can find pair of disjoint open neighborhood U_x, U_y with respect to x, y . We collect all of this into:

$$\mathcal{A} = \{U_y \mid y \in C\}$$

Then since this forms a open covering of C in X , we can apply 10.4 and obtain finite subcovering of C :

$$\mathcal{A}' = \{U_{y_1}, \dots, U_{y_n}\}$$

We have its pair elements $U_x^1, \dots, U_x^n$ and

$$U = \bigcap_{i \in [n]} U_{x_i} \implies \text{open neighborhood of } x \text{ disjoint with } C$$

So, $U \subseteq X \setminus C$, implies that $X \setminus C$ is open in X . □

We bring some useful features in above proof into following lemma. This will be appear in sections about separation axioms.

Lemma 10.7

If Y is a compact subspace of Hausdorff space X and $x \notin Y$, then there exists two disjoint open sets U and V :

$$U \text{ is neighborhood of } x, \quad Y \subseteq V$$

Next few theorems construct the connection between continuous map and homeomorphisms with compactness.

Theorem 10.8. Continuous Map Preserves Compactness

The image of compact space under continuous map is compact.

Proof. Straight forward, so omitted. □

Corollary 10.9. Compactness is Absolute

Let Y be the subspace of X . If K is compact in Y , then K also compact in X .

Proof. Natural inclusion $f : Y \hookrightarrow X$ is continuous. Since continuity preserves compactness, $f(K) = K \subseteq X$ also compact. □

Theorem 10.10. From Compact into Hausdorff

Let $f : X \rightarrow Y$ be a bijective continuous map. If X is compact and Y is Hausdorff then f is a homeomorphism.

Proof. Only thing we need to do is that show f^{-1} is continuous. Then it is enough to show that, image of closed set in X is also closed in Y . First, if $A \subseteq X$ is closed, then it is compact by 10.5. However, image of compact subspace under continuous map is also compact by previous theorem. Finally, compact subspace $\text{Im}(A)$ under f in Hausdorff space Y is closed by 10.6. $\square$

Now we prove some important and quite non-trivial statements.

Lemma 10.11. Tube Lemma

Let $X \times Y$ be a product space where Y is compact. If N is open set of $X \times Y$ containing $x_0 \times Y$ for some $x_0 \in X$, then N contains some *tube* $W \times Y$ about $x_0 \times Y$, where W is a neighborhood of x_0 in X .

Proof. Since N is an open set of $X \times Y$, we can denote N is a union of basis element $\mathcal{T}_X \times \mathcal{T}_Y$. So, denote

$$N = \bigcup_{\alpha \in J} (U_\alpha \times V_\alpha), \quad U_\alpha \in \mathcal{T}_X, \quad V_\alpha \in \mathcal{T}_Y$$

Now here, since $x_0 \times Y \subseteq N$, $\{V_\alpha \mid \alpha \in J, x_0 \in U_\alpha\}$ is open covering of Y and so we can find finite subcovering of Y by

$$\{V_i \mid i \in [n], x_0 \in U_i\} \text{ such that } \bigcup_{i \in [n]} V_i = Y$$

Then, we construct

$$W = \bigcap_{i \in [n]} U_i \implies W \text{ is neighborhood of } x_0 \text{ in } X$$

Then it is clear that $W \times Y \subseteq N$. $\square$

Theorem 10.12. Product of Finite Compact is Compact

The product of finitely many compact space is compact.

Proof. It is sufficient to show that if X, Y is compact then $X \times Y$ is compact. Let $\mathcal{A}$ be any open covering of $X \times Y$. Then for any fixed $x_0 \in X$, $x_0 \times Y$ is homeomorphic to Y and so compact. By 10.4, $x_0 \times Y$ is compact subspace of $X \times Y$ and $\mathcal{A}$ is a open covering of $x_0 \times Y$, so we can find finite subcovering $A_1, \dots, A_n \in \mathcal{A}$ such that $x_0 \times Y \subseteq \bigcup A_i = N$. Then according to 10.11, we can find $W_{x_0} \times Y \subseteq N$ such that W is neighborhood of x_0 in X . Do this for every $x_0 \in X$ then collect

$$\{W_x \mid x \in X\} \implies \text{open covering of } X \implies \text{Finite subcovering } \{W_{x_1}, \dots, W_{x_m}\}$$

Then,

$$\{W_{x_i} \times Y \mid i \in [m]\} \implies \text{Finite Subcovering of } \mathcal{A} \text{ of } X \times Y$$

This completes the proof. $\square$

Theorem 10.13. Tychonoff Theorem

Product of arbitrary many compact spaces is compact.

Proof. We prove this later. $\square$

Finally, we propose another criterion for space to be compact.

Definition 10.14. Finite Intersection Property

A collection $\mathcal{C}$ of subsets of X is said to have the *finite intersection property* if for every finite subcollection

$$\{C_1, \dots, C_n\}$$

of $\mathcal{C}$, the intersection $C_1 \cap \dots \cap C_n$ is nonempty.

Theorem 10.15. Finite Intersection Property Criterion

X is compact if and only if every collection $\mathcal{C}$ of *closed subsets* in X having finite intersection property. i.e., $\bigcap_{C \in \mathcal{C}} C \neq \emptyset$.

Proof. First, it is important to observe the connection between definition of open covering and finite intersection property of closed sets. Let $\mathcal{C}$ be a given collection of closed sets in X . Then, we can construct

$$\mathcal{A} = \{X \setminus C \mid C \in \mathcal{C}\}$$

This is collection of open sets in X . Suppose that $\mathcal{A}$ has a finite subcovering of X .

$$\{A_1, \dots, A_n\} \subseteq \mathcal{A} \implies \bigcup_{i \in [n]} A_i = X$$

This implies that

$$\bigcap_{i \in [n]} C_i = \emptyset \text{ where } C_i = X \setminus A_i \in \mathcal{C}$$

However, X is compact if and only if for any open covering $\mathcal{A}$, we can find finite subcovering in $\mathcal{A}$. So, X is compact if and only if any collection of open sets $\mathcal{A}$ doesn't have finite subcovering, then $\mathcal{A}$ not cover X . Latter can be translated into words of $\mathcal{C}$ and finite intersection property:

Any collection $\mathcal{C}$ of closed subsets in X having finite intersection property.

$\square$

11 Metric and Metric Topology

Before moving on to the compactness of Euclidean spaces, it is good time to define the general notion of distance, which is called *metric* and topology induced by it. $\mathbb{R}$ is very intuitive space for us and we learned so far about $\mathbb{R}$. In many theories including topology, sometimes they behaves like *meta-theory* of our domain. The notion of *metric* is sort of that. The *metric*, general notion of distance is defined as following.

Definition 11.1. Metric

A *metric* on a set X is a function

$$d : X \times X \rightarrow \mathbb{R}$$

with following axioms:

1. $d(x, y) \geq 0$ for every $x, y \in X$ and $d(x, y) = 0 \iff x = y$.
2. $d(x, y) = d(y, x)$ for any $x, y \in X$.
3. $d(x, y) + d(y, z) \geq d(x, z)$ for any $x, y, z \in X$.

Notation 11.2. Open Ball

Given metric d of X and $\varepsilon > 0$, we denote

$$B_d(x, \varepsilon) = \{y \mid d(x, y) < \varepsilon\}$$

and it is called ε -ball centered at x . When d is trivial in context, we denote $B(x, \varepsilon)$ often.

This two notion is very familiar from real analysis course. However, note that metric is defined on the arbitrary set X . So one natural further step is that, when metric on a set X is given, what is the natural topology for X induced by the metric?

Definition 11.3. Metric Topology

Let d be a metric on a set X . Topology generated by *collection of all ε -balls in X* is called the *metric topology induced by d* . i.e.,

$$\mathcal{B} = \{B_d(x, \varepsilon) \mid x \in X, \varepsilon > 0\}$$

This is well-defined since if $x \in B_1 \cap B_2$ where $B_1 = B(y, \varepsilon_1)$, $B_2 = B(z, \varepsilon_2)$ then we can find $\delta = \min(\varepsilon_1 - d(x, y), \varepsilon_2 - d(x, z))$ then $x \in B(x, \delta) \subseteq B_1 \cap B_2$.

Again also natural question is occurring. Since we show that given metric can induce natural topology of X , we want to ask the reverse:

When $(X, \mathcal{T})$ is given, can we find proper metric d that induces the topology of X ?

It's a sort of important question, and indeed, very strong condition in topology.

Definition 11.4. Metrizable

Topological space X is said to be *metrizable* if there exists a metric $d : X \times X \rightarrow \mathbb{R}$ such that induces the topology of X . A *metric space* is a metrizable space X together with a explicit metric d that induces that topology of X .

Once we can prove that X is metrizable, then it is *almost visualizable*, i.e., it becomes very intuitive and nature space for us. So many intuitive lemmata guaranteed on metrizable space, and also proving X is metrizable is important problem. Since they are main branch of topological theory, we'll see that in later section. Now we'll do almost recap from analysis course.

Definition 11.5. Bounded, Diameter

X is a metric space with metric d . $A \subseteq X$ is said to be bounded if there exists $M \in \mathbb{R}$ such that

$$\forall a_1, a_2 \in A, \quad d(a_1, a_2) \leq M$$

If A is bounded and nonempty, we can define *diameter* of A by

$$\text{diam}(A) = \sup\{d(a_1, a_2) \mid a_1, a_2 \in A\}$$

Theorem 11.6. Standard Bounded Metric

X is a metric space with metric d . We define $\bar{d} : X \times X \rightarrow \mathbb{R}$ by

$$\bar{d} : (x, y) \mapsto \min\{d(x, y), 1\}$$

Then $\bar{d}$ is a metric and it induces the same topology as d . This is called *standard bounded metric* corresponding to d .

Definition 11.7. Norm, Euclidean Metric, Square Metric

Given $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$, we define *norm* of $\mathbf{x}$ by

$$\|\mathbf{x}\| = (x_1^2 + \dots + x_n^2)^{1/2}$$

We define the *Euclidean metric* d on $\mathbb{R}^n$ by

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$$

We define the *square metric* ρ by

$$\rho(\mathbf{x}, \mathbf{y}) = \max\{|x_1 - y_1|, \dots, |x_n - y_n|\}$$

Lemma 11.8. Comparison of Metrics

Let d, d' be two metrics on X and $\mathcal{T}, \mathcal{T}'$ be topologies they induce, respectively. $\mathcal{T} \subseteq \mathcal{T}'$ if and only if $\forall x \in X$ and $\forall \varepsilon > 0$, there exists $\delta > 0$ such that $B_{d'}(x, \delta) \subseteq B_d(x, \varepsilon)$.

Proof. $\mathcal{T} \subseteq \mathcal{T}'$ if and only if for any basis $x \in B$ of $\mathcal{T}$ we can find $B' \in \mathcal{B}'$ such that $x \in B' \subseteq B$. This directly same statements as desired. $\square$

Corollary 11.9. Euclidean Metric and Square Metric on $\mathbb{R}^n$

On $\mathbb{R}^n$, topology generated by Euclidean metric and square metric are same.

Now we have the cases what we've seen in section 6 which is arbitrary cartesian product space $\mathbb{R}^J$. When J is infinite index set, the definition of Euclidean metric and square metric is not always well-defined. So we now observe how can we generalize that cases and connect with box topology, product topology.

Definition 11.10. Uniform Metric, Uniform Topology

Let $\mathbf{x} = (x_\alpha)_{\alpha \in J}, \mathbf{y} = (y_\alpha)_{\alpha \in J}$ of $\mathbb{R}^J$. We define *uniform metric* $\bar{\rho}$ on $\mathbb{R}^J$ by

$$\bar{\rho}(\mathbf{x}, \mathbf{y}) = \sup\{\bar{d}(x_\alpha, y_\alpha) \mid \alpha \in J\}$$

where $\bar{d}$ is the standard bounded metric on $\mathbb{R}$ in 11.6. Topology induced by it is called the *uniform topology* of $\mathbb{R}^J$.

Theorem 11.11. Product $<$ Uniform $<$ Box in $\mathbb{R}^J$

In $\mathbb{R}^J$,

$$\text{Product topology} \subseteq \text{Uniform topology} \subseteq \text{Box topology}$$

If J is infinite,

$$\text{Product topology} \subset \text{Uniform topology} \subset \text{Box topology}$$

Proof. Let $\mathbf{x} = (x_\alpha)_{\alpha \in J}$ and $\mathbf{x} \in \prod U_\alpha$ which is a basis of product topology. So, U_α is non- $\mathbb{R}$ open subset of $\mathbb{R}$ in $\alpha_1, \dots, \alpha_n$ and $U_\alpha = \mathbb{R}$ for other indices. Since U_{α_i} is open in $\mathbb{R}$ and $x_{\alpha_i} \in U_{\alpha_i}$, we can find ε_i such that $B(x_{\alpha_i}, \varepsilon_i) \subseteq U_{\alpha_i}$. Since standard bounded metric induces same topology of $\mathbb{R}$ as Euclidean metric, we can denote

$$B_{\bar{d}}(x_{\alpha_i}, \varepsilon_i) \subseteq U_{\alpha_i}$$

Let $\varepsilon = \min\{\varepsilon_1, \dots, \varepsilon_n\} > 0$. Then we can define $B_{\bar{\rho}}(\mathbf{x}, \varepsilon)$. It is clear that

$$\mathbf{x} \in B_{\bar{\rho}}(\mathbf{x}, \varepsilon) \subseteq \prod_{\alpha \in J} U_\alpha$$

So, uniform topology is finer than product topology.

Let $\mathbf{x} = (x_\alpha)_{\alpha \in J}$ and $\mathbf{x} \in B_{\bar{\rho}}(\mathbf{x}, \varepsilon)$. Then we can construct

$$U_\alpha = \left(x_\alpha - \frac{1}{2}\varepsilon, x_\alpha + \frac{1}{2}\varepsilon \right)$$

for each $\alpha \in J$. Then, $\prod U_\alpha$ is basis of box topology such that

$$\mathbf{x} \in \prod U_\alpha \in B_{\bar{\rho}}(\mathbf{x}, \varepsilon)$$

□

Theorem 11.12. Product Topology is Metrizable

Let $\bar{d}$ be the standard bounded metric on $\mathbb{R}$ referred in 11.6. Define

$$D : (\mathbf{x}, \mathbf{y}) \mapsto \sup \left\{ \frac{\bar{d}(x_i, y_i)}{i} \right\}$$

where $\mathbf{x}, \mathbf{y} \in \mathbb{R}^\omega$, set of sequences in $\mathbb{R}$. (i.e., countable infinite) Then D is a metric that induces the product topology on $\mathbb{R}^\omega$.

Proof. First, we claim that D is a metric of $\mathbb{R}^\omega$. The metric axioms in 11.1 (1), (2) is obvious. For any $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^\omega$, since $\bar{d}$ is a metric of $\mathbb{R}$,

$$\bar{d}(x_i, z_i) \leq \bar{d}(x_i, y_i) + \bar{d}(y_i, z_i) \implies \frac{\bar{d}(x_i, z_i)}{i} \leq \frac{\bar{d}(x_i, y_i)}{i} + \frac{\bar{d}(y_i, z_i)}{i} \leq D(\mathbf{x}, \mathbf{y}) + D(\mathbf{y}, \mathbf{z})$$

for every i . So,

$$D(\mathbf{x}, \mathbf{z}) \leq D(\mathbf{x}, \mathbf{y}) + D(\mathbf{y}, \mathbf{z})$$

So D is a metric of $\mathbb{R}^\omega$.

Now we prove that topology induced by D is same with product topology of $\mathbb{R}^\omega$. Let $\mathbf{x} = (x_\alpha)_{\alpha \in J}$. First, we choose a basis of topology induced by D :

$$B_D(\mathbf{x}, \varepsilon) \implies \sup \left\{ \frac{\bar{d}(x_i, y_i)}{i} \right\} < \varepsilon$$

Then, there exists $m \in \mathbb{N}$ such that $\varepsilon \cdot m > 1$. So,

$$i \geq m \implies \bar{d}(x_i, y_i) \leq 1 \implies y_i \in \mathbb{R}$$

Then we can construct

$$\prod U_i, \quad U_i = \begin{cases} B(x_i, \varepsilon) & i < m \\ \mathbb{R} & i \geq m \end{cases}$$

Then for $\mathbf{y} \in \prod U_i$,

$$D(\mathbf{x}, \mathbf{y}) = \max_{1 \leq i < m} \left\{ \frac{\bar{d}(x_i, y_i)}{i}, \frac{1}{m} \right\} < \varepsilon$$

So, $\mathbf{x} \in \prod U_i \subseteq B_D(\mathbf{x}, \varepsilon)$.

Second, we choose a basis of product topology:

$$\mathbf{x} \in \prod U_i, \quad U_i = \begin{cases} \text{Open set of } \mathbb{R} \text{ that } \neq \mathbb{R} & U_{i_1}, \dots, U_{i_n} \\ \mathbb{R} & \text{otherwise} \end{cases}$$

Then, we can find $0 < \varepsilon_1, \dots, \varepsilon_n < 1$ such that

$$B_d(x_{i_j}, \varepsilon_j) \subseteq U_{i_j}, \quad j \in [n]$$

Now, let

$$\varepsilon = \min \left\{ \frac{\varepsilon_1}{i_1}, \dots, \frac{\varepsilon_n}{i_n} \right\}$$

Then, $\mathbf{x} \in B_D(\mathbf{x}, \varepsilon) \subseteq \prod U_i$ □

Theorem 11.13. Subspaces of Metric Space is Metric Space

Let X be a metric space with metric d . If A is subspace of X with subspace topology, $d|_{A \times A}$ is metric of A and it induce that topology.

Lemma 11.14. Metric Space is Hausdorff

Metric space is Hausdorff.

Once metric is defined, the notion of continuity become more intuitive to us. As I mentioned, the formal definition of continuity said that ‘near images are from near domain’. The concept ‘near’ can be dealt with numerical number in metric space. ‘ ε -near images are from δ -near domain elements’.

Theorem 11.15. Continuity on Metrizable Space

Let $f : X \rightarrow Y$ and X, Y are metrizable with d_X, d_Y respectively. The continuity of f is equivalent to that given any $x \in X$ and $\varepsilon > 0$, there exists $\delta > 0$ such that

$$d_X(x, y) < \delta \implies d_Y(f(x), f(y)) < \varepsilon$$

In analysis, the sequence is sort of important feature we studied. We might want to generalize the concept of sequences and convergence also. First one is the relation between limit points and convergent sequence.

Lemma 11.16. The Sequence Lemma

Let X be a topological space. For $A \subseteq X$, if there is a sequence of points of A converging to x , then $x \in \bar{A}$. The converse holds if X is metrizable. (i.e., if X is metrizable there exists a sequence of points of A such that converges into $x \in \bar{A}$.)

Proof. Let $\{a_n\}_{n \in \mathbb{N}}$ converges to $x \in X$. By the definition, for any neighborhood U of x in X , there exists $N \in \mathbb{N}$ such that if $a_n, n \geq N$ then $a_n \in U$. This automatically implies that for any neighborhood U of x , $U \cap A \neq \emptyset$ so $x \in \bar{A}$.

Let X be metrizable with d and $x \in \bar{A}$. Then, we can construct

$$\{B_d(x, 1/n) \mid n \in \mathbb{N}\}$$

becomes sequence of neighborhood of x in X . However since $x \in \bar{A}$, we can find elements of A in each open balls. Let's denote them by $\{a_n\}_{n \in \mathbb{N}}$. It converges to x in A . $\square$

Theorem 11.17. Continuity Preserves Convergence

If $f : X \rightarrow Y$ is continuous and $\{x_n\}_{n \in \mathbb{N}} \rightarrow x$ is a convergent sequence in X then $\{f(x_n)\}_{n \in \mathbb{N}}$ converges to $f(x)$ in Y . The converges holds if X is metrizable.

Proof. The preimage of neighborhood U of $f(x)$ in Y is neighborhood V of x in X , so we can find $N \in \mathbb{N}$ such that $x_n \in V$ for $n \geq N$. So, all of images of them send into U , so $\{f(x_n)\}$ converges to $f(x)$.

If X is metrizable and latter condition holds, we'll see the continuity of f by $f(\bar{A}) \subseteq \overline{f(A)}$. If $x \in \bar{A}$, then we can find $\{x_n\}$ in A converges to x since X is metrizable, by 11.16. Then, $\{f(x_n)\}$ converges to $f(x)$ by the condition. So, $f(x) \in \overline{f(A)}$. This implies that $f(\bar{A}) \subseteq \overline{f(A)}$, f is continuous by 5.5. $\square$

As you might realized, we used metrizable property while proving 11.16 however 'not very much' use the power of metrizable spaces. We just used to obtain **sequentially decreasing neighborhood of x** . So indeed, 11.16 and 11.17 holds on weaker condition. It is quite non-trivial definition, but we'll revisit this concept in latter section.

Definition 11.18. Countable Basis, First Countability

A space X is said to have a *countable basis at the point x* if there is a countable collection $\{U_n\}_{n \in \mathbb{N}}$ of neighborhoods of x such that any neighborhood U of x contains at least one of the sets U_n . i.e., $\forall U$ neighborhood of x , we can find U_n such that $x \in U_n \subseteq U$ from countable many $\{U_n\}$. If X has a countable basis at each points then X satisfy the *first countability axiom*.

Remark 11.19. Holds on First Countable Space X

We can replace $B(x, 1/n)$ into $B_n = U_1 \cap \cdots \cap U_n$ for countable basis $\{U_n\}$ of x in proof of 11.16 and so on 11.17.

In analysis, we also dealt with sequence of functions and its convergences.

Definition 11.20. Uniform Convergence

Let $\{f_n : X \rightarrow Y\}$ be a sequence of functions from set X to metric space Y with d . We say that $\{f_n\}$ *converges uniformly* to f and denote $\{f_n\} \Rightarrow f$ if given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$d(f_n(x), f(x)) < \varepsilon$$

for all $n > N$ and $\forall x \in X$.

Theorem 11.21. Uniform Limit Theorem

Let $f_n : X \rightarrow Y$ be a sequence of continuous functions from topological space X into metric space Y . If $\{f_n\}$ uniformly converges to f then f is continuous.

Proof. Let V be open in Y . For $y_0 \in V$, let $f(x_0) = y_0$. Then we can find $B(y_0, \varepsilon) \subseteq V$. Since $\{f_n\} \Rightarrow f$, we can find $N \in \mathbb{N}$ such that $d(f_n(x), f(x)) < \varepsilon/3$ for any $x \in X$. Here, f_N is continuous map, so we can state that $U = f_N^{-1}(B(f_N(x_0), \varepsilon/3))$ is open neighborhood of x_0 in X . Then for $x \in U$,

$$d(f(x), f_N(x)) < \varepsilon/3 \quad \because \text{uniform continuity}$$

$$d(f_N(x), f_N(x_0)) < \varepsilon/3 \quad \because x, x_0 \in U$$

$$d(f_N(x_0), f(x_0)) < \varepsilon/3 \quad \because \text{uniform continuity}$$

This implies that if $x \in U$, $d(f(x), f(x_0)) < \varepsilon$ so $x \in U$ implies that $U \subseteq f^{-1}(V)$, so $f^{-1}(V)$ is open in X . $\square$

Remark 11.22. Uniform Metric and Uniform Convergence

The *uniform metric* is related with *uniform convergence*.

Recap, uniform metric on $\mathbb{R}^J$ is defined by

$$\bar{\rho} : (\mathbf{x}, \mathbf{y}) \mapsto \sup \{ \bar{d}(x_\alpha, y_\alpha) \mid \alpha \in J \}$$

For a set X , view $\mathbb{R}^X$ as collection of $f : X \rightarrow \mathbb{R}$ and give $\mathbb{R}^X$ to uniform metric. Suppose that $\{f_n\}$, the sequence of elements in $\mathbb{R}^X$ converges to f with respect to uniform metric. We can say that $f_n : X \rightarrow \mathbb{R}$ uniformly converges to f .

Remark 11.23. Not metrizable

To show that some space is not metrizable, in many case we show that such space does not holds sequence lemma with some counter example.

12 Compactness with Metric

Again, the motive of compact space is generally good space to derive. For motivation, we state following theorem without proof.

Theorem 12.1. Compact = Closed and Bounded in $\mathbb{R}^n$

$A \subseteq \mathbb{R}^n$ is compact if and only if A is closed and bounded in $\mathbb{R}^n$ under Euclidean metric or square metric.

Anyone read this theorem might agree that such space is intuitively, enough good for analysis. One good property of compact space is stated in following theorem.

Theorem 12.2. Extreme Value Theorem

Let $f : X \rightarrow Y$ be continuous function where Y is order topology space. If X is compact, then $\exists c, d \in X$ such that $f(c) \leq f(x) \leq f(d)$ for any $x \in X$.

Proof. Since continuous map preserves compactness by 10.8, $f(X) \subseteq Y$ is compact. So, since

$$\{(-\infty, a) \mid a \in f(X)\} \implies \text{open cover of } f(X)$$

we can find finite subcovering,

$$\{(-\infty, a_1), \dots, (-\infty, a_n)\} \implies \max\{a_i\} = \sup f(X) = f(d)$$

Similarly for $\inf f(X)$. □

From now on, *the most important and useful* theorems will be stated. Before moving on, we define one more very intuitive notion: distance from point-to-set. There are few way to define it, however following definition is the most standard one.

Definition 12.3. Distance from Point to Set

Let (X, d) be a metric space and $\emptyset \neq A \subseteq X$. For $x \in X$, we can define the *distance* from x to A by

$$d(x, A) := \inf\{d(x, a) \mid a \in A\}$$

Remark 12.4. Distance is Continuous

For fixed A , distance is continuous over X .

Now here we state one of the most important theorem in this summary note.

Theorem 12.5. Lebesgue Number Lemma

Let $\mathcal{A}$ be an open covering of the metric space (X, d) . If X is compact then there exists $\delta > 0$ such that each subset of X having diameter less than δ , there exists an element of $\mathcal{A}$ containing it. Such δ is called a **Lebesgue number** for the covering $\mathcal{A}$.

Proof. Let $\mathcal{A}$ be any open covering of X . If $X \in \mathcal{A}$, then statement automatically holds for any $\delta > 0$. So suppose that $X \notin \mathcal{A}$. Since X is compact, we can find finite subcovering $\{A_1, \dots, A_n\} \subseteq \mathcal{A}$. Denote $C_i = X \setminus A_i$ for each $i \in [n]$. Define

$$f(x) = \frac{1}{n} \sum_{i=1}^n d(x, C_i)$$

Then $f(x) > 0$ (strictly larger than 0) since x must contained in some open cover ($= A_i$), so

$$x \in A_i \implies B_d(x, \varepsilon) \subseteq A_i \implies d(x, C_i) > \varepsilon \implies f(x) > 0$$

However, according to 12.4 and summation of continuous functions are continuous, f is continuous over X . Since X is compact, $f(X)$ also compact by 10.8 and so closed and bounded. We can find minimum($= \delta$) of $f(X)$. This $\delta > 0$ become a Lebesgue number we claimed. Suppose that $B \subseteq X$ and diameter of $B < \delta$. Let $x_0 \in B$ then

$$\delta \leq f(x_0) \leq d(x_0, C_m) \text{ where } C_m \text{ is the most far } C_i$$

Implies that $B(f(x_0), \delta) \subseteq A_m \in \mathcal{A}$. □

Now view $\mathcal{A}$ as the *collection of open sets* such that as many as enough to cover X totally. The Lebesgue number theorem states that, if X is compact space then we can assume that any sufficiently small(δ -small) subset of X is totally contained in some given open set.

In compact space another good property also arises which is called *uniform continuous*.

Definition 12.6. Uniform Continuous

$f : X \rightarrow Y$ is a continuous map where X, Y are metric space induced by d_X, d_Y respectively. f is uniformly continuous if for any $\varepsilon > 0$, we can find $\delta > 0$ such that

$$d_X(x_0, x_1) < \delta \implies d_Y(f(x_0), f(x_1)) < \varepsilon$$

We describe that the continuity intuitively implies that *near elements(neighborhood) of codomain is from a near elements(neighborhood) of domain*. Uniform continuity (1) Explicitly determines the ‘near’ via metric d_X, d_Y respectively. (2) Such nearness is ‘globally determined’ for X, Y both.

Theorem 12.7. Uniform Continuity Theorem

Continuous function from compact metric space into general metric space is uniformly continuous.

Proof. Easily proved by 12.5. Find δ from preimage of $\varepsilon/2$ -open balls for each element of Y . For $x_0 \in X$, image of any $\delta/2$ -ball centered by x_0 is contained in $\varepsilon/2$ -ball. $\square$

Finally we prove that the uncountability of real number. We say that $x \in X$ is an *isolated point* of X if $\{x\}$ is open in X .

Theorem 12.8. Uncountability of Compact Hausdorff with Non-isolated

Let X be a nonempty compact Hausdorff space. If X has no isolated points, then X is uncountable.

Proof. Let $x \in X$ and U be arbitrary chosen nonempty open set of X . If $x \in U$, since x is not isolated point, $\{x\} \subset U$ so we can find $x \neq y \in U$. If $x \notin U$, since U is nonempty we can find $x \neq y \in U$. Since X is Hausdorff, we can construct two open sets W_x, W_y distinguishing x, y respectively. Let $V = W_y \cap U$. Again it distinguishes x in U and open. Since $x \in W_x \subseteq X \setminus V$, we can say that $x \notin \bar{V}$.

Let $f : \mathbb{N} \rightarrow X$ and $f(n) := x_n$. Let $V_0 = X$. We can find $V_1 \subset V_0$ such that $x_1 \notin \bar{V}_1$. Similarly, we can find $V_n \subset V_{n-1}$ such that $x_n \notin \bar{V}_{n-1}$. Then now we obtain

$$X = \bar{V}_0 \supset \bar{V}_1 \supset \dots$$

Note that X is compact and each $\bar{V}_i$ is closed. by 10.14,

$$\exists x \in \bigcap_{i \in \mathbb{N}} \bar{V}_i$$

If $x = x_m$ for some $m \in \mathbb{N}$, $x_m \in \bar{V}_m$ but it is impossible. So for any given countable sequence, we can find $x \in X$ that not contained in it. So X is uncountable. $\square$

13 Limit Point Compactness

At the beginning of section 10, we mentioned that ‘compact’ means good-property inherited. We learned the standard version definition of such good property via finite subcovering exists in any given open covering. There are another version of ‘good’ which is called *limit point compactness*. This is generally weaker than standard compactness, however, they agree on the metrizable spaces.

Definition 13.1. Limit Point Compactness

A space X is said to be *limit point compact* if every infinite subset of X has a limit point.

This is quite natural. Once our space is limit point compactness, we can imagine that our space is sort of ‘bounded’ in some sence. Actually, general compactness also states the general concepts of bounded space.

Theorem 13.2. Compactness $\implies$ Limit Point Compactness

Compactness implies limit point compactness, but not conversely.

Proof. Suppose that X is compact but not limit point compact. Then, we can find $A \subseteq X$ infinite order but A doesn’t have any limit point. Then by definition, for all $a \in A$ we can find open neighborhood U_a of a such that $U_a \cap A = \{a\}$. $\mathcal{C} = \{U_a \mid a \in A\}$ be a open covering of A . However, X is compact so we can find $U_1, \dots, U_n$ that finite subcovering of A . But it cover only n elements of A , here contradiction occurs.

The minimal uncountable well-ordered set S_Ω is not compact but limit point compact. □

Similarly we define one more compactness which is called *sequentially compactness*.

Definition 13.3. Sequentially Compact

Let X be a topological space and $\{x_n\}$ is a sequence of points of X . Let

$$n_1 < n_2 < \dots < n_i < \dots$$

For sequence $\{y_i\}$, $y_i = x_{n_i}$ is called a subsequence of $\{x_n\}$. X is said to be *sequentially compact* if every subsequence of points of X has a convergent subsequence.

Theorem 13.4. Three Compacts are Same on Metrizable Space

On the metrizable space X ,

$$X \text{ is compact} \iff X \text{ is limit point compact} \iff X \text{ is sequentially compact}$$

Proof. Compactness implies limit point compactness for general X by 13.2.

(2) $\implies$ (3) Suppose that X is limit point compact. Let $\{x_n\}$ be any sequence of points of X . If $\{x_n\}$ is finite, then it already converges. Then if $\{x_n\}$ is infinite set over X , there exists a limit point x . We can find subsequence of $\{x_n\}$ such that converges to x by following algorithm:

$$y_1 \implies x_{n_1} \in B(x, 1)$$

Then,

$$y_i \implies x_{n_i} \in B(x, 1/i), n_{i-1} < n_i$$

Second process is well defined since there are infinitely many elements in $B(x, 1/i)$ and so we can always find such n_i .

(3) $\implies$ (1) Suppose that X is sequentially compact space. First, **Lebesgue number lemma holds on X** . Suppose that $\mathcal{A}$ be any open covering of X . Suppose that $\nexists \delta > 0$ such that $\text{diam}(B) < \delta \implies \exists A \in \mathcal{A}$ such that $B \subseteq A$ holds. Since such δ not exists, once we choose n , there exists a set diameter is less than $1/n$ but not contained in any element of $\mathcal{A}$ totally. Let C_n be such a set ($\text{diam}(C_n) < 1/n$ but C_n not totally contained in any $A \in \mathcal{A}$). For each C_n , choose $x_n \in C_n$. Then there exists $\{y_i\}, y_i = x_{n_i}$ that converges to a . Since $\mathcal{A}$ is a covering, $a \in A \in \mathcal{A}$ for some A . A is open set and $a \in A$, so we can find $B(a, \varepsilon) \subseteq A$. Since $\{x_{n_i}\} \rightarrow a$, we can find x_{n_i} such that $B(x_{n_i}, 1/n_i) \subseteq B(a, \varepsilon) \subseteq A$. However, this is contradiction. ($\because x_{n_i}$ is sampled from non- A -containing set size less than $1/n_i$)

Second, **given $\varepsilon > 0$, there exists finite covering of X by open ε -balls in X** . Suppose not, means that for some $\varepsilon > 0$, X can't be covered by finite ε -balls. Now we sample points in each ε -balls at most one. In general, given $x_1, \dots, x_n$, choose x_{n+1} to be a point not in the union

$$B(x_1, \varepsilon) \cup \dots \cup B(x_n, \varepsilon)$$

Then $\{x_n\}$ can not have convergent subsequence.

Finally, **X is compact**. Given any open covering $\mathcal{A}$ of X , we can find Lebesgue number δ . Let $\varepsilon = \delta/3$ and find finite covering of X by open ε -balls. Each of balls has size $< 2\delta/3$, so it lies in some $A \in \mathcal{A}$. Collect such A , it becomes finite subcovering of X from $\mathcal{A}$. So X is compact. $\square$

14 Local Compactness

Until now, when we say ‘compactness’ that was sort of global property of spaces(or subspaces). But as you did in open set, if we can choose *compact boundary containing* x in a (not necessarily compact) space, it can be also a good property. Indeed, $\mathbb{R}^n$ is sort of such space. Since all intuitions came from standard space like $\mathbb{R}^n$, we give name for such property.

Definition 14.1. Local Compactness

A space X is said to be *locally compact* at x if there exists compact subspace $C \subseteq X$ that contains a neighborhood of x . If X is locally compact at each of its points, X is said to be *locally compact space*.

Remark 14.2. Compact $\implies$ Locally Compact

Compact space is locally compact.

Proof. $\forall x \in X$, we can choose any neighborhood U of x in X . Then, $x \in U \subseteq \bar{U}$ is closed subset in compact space X , so X is locally compact. $\square$

Warning 14.3. Non Locally Compact Spaces

$\mathbb{Q}, \mathbb{R}^\omega$ (with product topology) is not locally compact.

From now on, we'll study *compactifications*. There are very good intuition about this in p.183 of [2]. As just quoting them, two of the most well-behaved classes of spaces for mathematics are the metrizable spaces and the *compact Hausdorff* spaces. If a given space is not of one types, the next best thing one can hope for is that it is a **subspace of one of those spaces since we can imbedding**. However, subspace of a compact Hausdorff space need not be compact. **Then, under what conditions is a space homemomorphic with a subspace of a compact Hausdorff space?**

Theorem 14.4. Compactification Theorem

X is *locally compact Hausdorff* if and only if there exists a space Y satisfying the following conditions:

1. X is a subspace of Y
2. $Y \setminus X$ is consists of a single point.
3. Y is a compact Hausdorff space.

Such Y is unique up to homeomorphisms which is the identity map on X .

Proof. Suppose that X is locally compact Hausdorff space. Let $\infty \notin X$ be the newly introduced object. Let's construct

$$Y = X \cup \{\infty\}$$

And we give following topology for Y .

$$\mathcal{T}_Y = \mathcal{T}_X \cup \{Y \setminus C \mid C \text{ is compact in } X\}$$

This satisfies the axiom of topology:

1. $\emptyset \in \mathcal{T}_X$, $Y = Y \setminus \emptyset \in \{Y \setminus C\}$
2.
 - For $U_1, U_2 \in \mathcal{T}_X$, $U_1 \cap U_2 \in \mathcal{T}_X$.
 - For $U_1 \in \mathcal{T}_X$, $U_2 = Y \setminus C$, since $U_1 \subseteq X$, $U_1 \cap U_2 = U_1 \cap (X \setminus C)$. Since X is Hausdorff, compact subset C is closed so $X \setminus C$ is open. It implies that $U_1 \cap U_2 = U_1 \cap (X \setminus C) \in \mathcal{T}_X$.
 - For $U_1 = Y \setminus C_1$, $U_2 = Y \setminus C_2$, we obtain $U_1 \cap U_2 = Y \setminus (C_1 \cup C_2)$. Then by 10.3, $C_1 \cup C_2$ is compact in X and so $U_1 \cap U_2 \in \{Y \setminus C\}$.
3.
 - Arbitrary union in $\mathcal{T}_X$ is in $\mathcal{T}_X$.
 - $\bigcup (Y \setminus C_\beta) = Y \setminus (\bigcap C_\beta)$, however by 10.3, in Hausdorff space arbitrary intersection of compact set is compact. So, it is in $\{Y \setminus C\}$.
 - $(\bigcup U_\alpha) \cup (\bigcup Y \setminus C_\beta) = U \cup (Y \setminus C) = Y \setminus (C \setminus U)$. However, $C \setminus U$ is closed subset of compact set C so $C \setminus U$ is compact, it is in $\{Y \setminus C\}$.

Then remaining things to do is showing that: (1) X is subspace of Y . (2) Y is compact. (3) Y is Hausdorff.

(1) If $U \in \mathcal{T}_X$, then $U \cap X = U \in \mathcal{T}_X$. If $U = Y \setminus C$, $U \cap X = X \setminus C \in \mathcal{T}_X$ since C is closed. And since $\mathcal{T}_X \subseteq \mathcal{T}_Y$, X is subspace of Y .

(2) Let $\mathcal{A}$ be an open covering of Y . To cover $\infty \in Y$, $\mathcal{A}$ must contain at least one open set of type $A = Y \setminus C$. Then, $\mathcal{A} - A$ is open covering of C . Since C is compact in X and by 10.9 it also compact in Y . So there are finite subcovering $\{A_n\}$ from $\mathcal{A} - A$, then $\{A_n\} \cup A$ becomes finite subcovering of $\mathcal{A}$ of Y .

(3) If $x, y \in X \subseteq Y$, then we can separate them via open sets. Without loss of generality, let $x \in X$, $y = \infty$. Since X is locally compact, we can construct neighborhood U of x containing in C . Then, $y \in Y \setminus C$ which is open in Y . U and $Y \setminus C$ is open disjoint neighborhood of x, y respectively in Y .

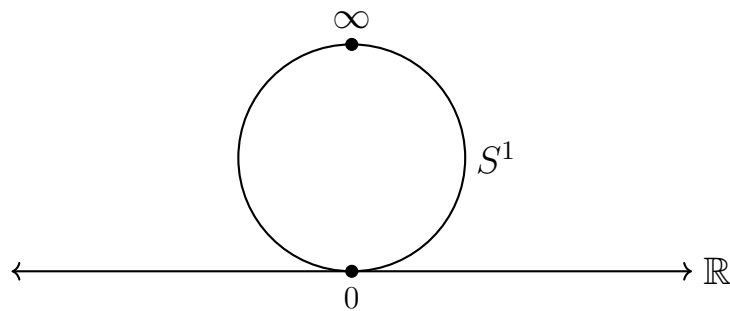
Now we prove converse. Suppose that Y satisfies conditions in the state. Then X is subspace of Hausdorff, so it also become Hausdorff. Let $x \in X$. Since $Y \setminus X = \{\infty\}$

by second condition and Y is Hausdorff, in Hausdorff space Y , we can find disjoint neighborhood U, V of x, ∞ respectively. Then $Y \setminus V$ is closed in Y and contained in X . Since Y is compact, $Y \setminus V$ is compact by 10.5. By 10.4, it is a compact in X and containing U . So, X is locally compact.

Uniqueness of embedding is obvious. $\square$

This is the notion of compactification. More formally, we say that construction of Y in 14.4 is *one-point compactification* of X . In other words, X is locally compact Hausdorff if and only if X has the unique one-point compactification up to homeomorphism.

Classical example is following. We denote that S^n by one-point compactification of $\mathbb{R}^n$.



15 Seperation Axioms

Have you ever considered the topological ‘resolution’ of a space? Our previous encounter with Hausdorff spaces illustrated that topology is deeply concerned with the ability to separate distinct points using open neighborhoods. In this section, we will explore this concept in full generality. The fundamental measure of a space’s ‘resolution’ is the degree to which its points and closed sets can be *topologically separated*. Much like how higher resolution provides greater clarity in optics, stronger topological conditions allow us to distinguish finer structural details within a space. We will formalize these levels of resolution through the hierarchy of Separation Axioms, categorizing spaces by what topological entities they can successfully separate.

Before moving on to the definition, all the concepts consisted with 2 parts: (1) What objects we’re trying to sepearate? (2) How we define such objects are seperated?

Definition 15.1. T_1 space

A space X is T_1 if for any pair of distinct points x, y , there exists an open set U with $x \in U, y \notin U$.

Under this basic resolution, we can state following.

Proposition 15.2. $T_1 \iff$ Singleton Set is Closed

X is T_1 if and only if every singleton $\{x\}$ is a closed set.

Proof. If X is T_1 , for any point y in $X - x$, we can find $y \in U, x \notin U$. So, $y \in U \subseteq X - x$, implies that $X - x$ is open, so $\{x\}$ is closed.

If any $\{x\}$ is closed set, then with similar reasoning, we can seperated x, y by open set such that $x \in U, y \notin U$. □

Definition 15.3. T_2 space, Hausdorff space

A space X is T_2 (Hausdorff) if for every pair of distinct points x, y there exists disjoint open sets U, V with $x \in U, y \in V$.

Proposition 15.4. Uniqueness of Limits

In a Hausdorff space, every sequence converges to at most one point.

Definition 15.5. T_3 space, Regular space

A space X is T_3 (Regular) if it is T_1 and for any $x \in X$ and every closed set C such that $x \notin C$, there exists disjoint open sets U, V such that $x \in U, C \subseteq V$.

Note that seperating object is upgraded from T_2 . (point/point $\rightarrow$ point/closed set).

Proposition 15.6. Equivalent Characterization of Regularity

A T_1 space X is regular if and only if $\forall x \in X$ and every open W containing x , there exists open neighborhood U of x with

$$x \in U \subseteq \bar{U} \subseteq W$$

Proof. Let X be a regular space. For arbitrary given x and open W that $x \in W$, x and $X \setminus W$ are separable via open sets. So,

$$U, V \text{ are disjoint open sets such that } x \in U, X \setminus W \subseteq V$$

Means that $x \in U \subseteq X \setminus V$. Since $X \setminus V$ is closed,

$$x \in U \subseteq \bar{U} \subseteq W$$

Suppose that later condition holds. For arbitrary x and closed set C that $x \notin C$, $W = X \setminus C$ is open and containing x , so we can find open U such that $x \in U \subseteq \bar{U} \subseteq X \setminus C$. Then, let $V = X \setminus \bar{U}$. It is open and U, V are disjoint and $C \subseteq V$. $\square$

Definition 15.7. T_4 space, Normal space

A space X is T_4 (Normal) if it is T_1 and for every pair of disjoint closed set A, B , there exists disjoint open sets U, V with $A \subseteq U$ and $B \subseteq V$.

You can easily obtain following result.

$$T_4 \implies T_3 \implies T_2 \implies T_1$$

We upgrades the separated objects from T_1 to T_4 , with same separation scheme: via open set. So this can view as ‘resolution level’ of given space. One natural question is that what’s the resolution level of spaces we’ve learned before?

Theorem 15.8. Metrizable Space is T_4 (Normal)

Every metrizable space is Normal.

Proof. Let X be a metrizable space and d be a metric of X . Given any two disjoint closed sets A, B of X , define following function:

$$f(x) = \frac{d(x, A)}{d(x, A) + d(x, B)}$$

Since A, B are disjoint, this is well-defined(denominator is non-zero). Since distance is continuous, f is continuous and $f(A) = \{0\}, f(B) = \{1\}$. Then $U = f^{-1}([0, 1/2)), V = f^{-1}((1/2, 1])$ are disjoint open sets with $A \subseteq U, B \subseteq V$. $\square$

Theorem 15.9. Compact Hausdorff space is Normal

Every compact Hausdorff space is Normal.

Proof. Let X be compact Hausdorff space and A, B are two disjoint closed subset in X . Suppose that we choose any point $\alpha \in A$. For every point $\beta \in B$, since X is Hausdorff, we can find U_α^β, V_β such that $\alpha \in U_\alpha^\beta$ and $\beta \in V_\beta$. Collection of V_β such that $\{V_\beta \mid \beta \in B\}$ forms an open covering of B . Since B is closed subset in compact space, B also compact by 10.5. So we can find $\{V_{\beta_1}, \dots, V_{\beta_n}\}$ a finite subcovering. Then we have corresponding $U_\alpha^{\beta_i}$ an open neighborhood of α . Then,

$$U_\alpha^* = \bigcap U_\alpha^{\beta_i}, V_\alpha^* = \bigcup V_{\beta_i} \implies U_\alpha^* \cap V_\alpha^* = \emptyset, \text{ both are open, } \alpha \in U_\alpha^*, \quad B \subseteq V_\alpha^*$$

Do this all for $\alpha \in A$, obtaining $\{U_\alpha^* \mid \alpha \in A\}$ forms open covering of A . By the similar reasoning, we finally obtain finite subcovering $\{U_{\alpha_i}^*\}_{i=1}^m$

$$A \subseteq U = \bigcup U_{\alpha_i}^*, \quad B \subseteq V = \bigcap V_{\alpha_i}^*$$

Where U, V are disjoint open set in X . □

Example 15.10. Examples

The Sorgenfrey plane $\mathbb{R}_l \times \mathbb{R}_l$ is not normal although $\mathbb{R}_l$ is normal. $\mathbb{R}$ with K -topology is not regular.

Definition 15.11. Tychonoff Space

A space X is $T_{3.5}$ (Completely Regular, Tychonoff) space if it is T_1 and for every $x \in X$ and every closed set C with $x \notin C$, there exists a continuous function $f : X \rightarrow [0, 1]$ with $f(x) = 0$ and $f(C) = 1$.

In general,

$$T_4 \implies T_{3.5} \implies T_3 \implies T_2 \implies T_1$$

$T_{3.5} \implies T_3$ is clear but $T_4 \implies T_{3.5}$ need quite amount of jobs, which is called *Urysohn's Lemma*. In the next section, we'll mainly focus on Urysohn's lemma. Before moving on, we also characterize Normality condition in other words. i.e., there are always perfect gap between closed set and its open boundary.

Lemma 15.12. Shrinking Lemma, Characterization of Normal

A T_1 space X is normal if and only if for every closed set C and every open set W with $C \subseteq W$, there exists an open set V with $C \subseteq V \subseteq \bar{V} \subseteq W$.

Proof. Suppose that X is T_1 Normal space. Given arbitrary closed set C and open set W with $C \subseteq W$, we can separate C and $X \setminus W$ via open sets. i.e., $C \subseteq V, X \setminus W \subseteq U$

where U, V are disjoint open sets. However, $C \subseteq V \subseteq X \setminus U$ however $X \setminus U$ is closed. So we obtain

$$C \subseteq V \subseteq \bar{V} \subseteq W$$

Conversely, let A, B be disjoint closed sets in X . Then find open set U, V such that $A \subseteq U \subseteq \bar{U} \subseteq X \setminus B$, and vice versa for B and V . U, V are disjoint open sets such that separates A, B . □

16 More About Normal Spaces

Definition 16.1. First Countable

A space X is first countable if for any point $x \in X$, there exists countable open neighborhoods $\{U_n\}_{n \in \mathbb{N}}$ of x such that for arbitrary open set U containing x , we can find U_i such that $x \in U_i \subseteq U$.

Definition 16.2. Second Countable

A space X is *second countable* if it has a countable basis $\mathcal{B} = \{B_n\}$.

Definition 16.3. Lindelöf Space

A space X is Lindelöf if for every open covering $\mathcal{A}$ of X , we can find *countable* subcovering of $\mathcal{A}$.

Theorem 16.4. Regular + Second Countable $\implies$ Normal

Every second countable regular(T_3) space is normal(T_4).

Proof. Let $\mathcal{B} = \{L_n\}$ be the countable basis of X and A, B are disjoint closed sets in X . Since A, B are closed and X is regular, we can find open sets U, V such that

$$x \in V \subseteq \bar{V} \subseteq U \subseteq X \setminus B$$

By the definition of basis, we can find a basis $L_i \in \mathcal{B}$ such that $x \in L_i \subseteq V$. Do this for all $x \in A$ and collect L_i . Then such collection is subcollection of $\{L_n\}$, so countable and it becomes the *countable subcovering* of A of $\mathcal{B}$. Denote it by $\{U_n\}$. Do exactly same things for B and construct countable subcovering $\{V_n\}$ of B . Now, let $U = \bigcup U_n$ and $V = \bigcup V_n$ then they are open sets containing A, B respectively. Yet, they are not need to be disjoint. Let's newly define

$$U'_n = U_n \setminus \bigcup_{i=1}^n \bar{V}_i, \quad V'_n = V_n \setminus \bigcup_{i=1}^n \bar{U}_i$$

Then clearly U'_n, V'_n are open, and $\{U'_n\}$ covers A , $\{V'_n\}$ covers B . Finally,

$$U' = \bigcup U'_n, \quad V' = \bigcup V'_n$$

are disjoint open sets such that separating A, B in X . □

17 Urysohn's Metrization Theorem

All the notions we defined in Section 15 are highly *topological*. However, they are clear enough for us to *visualize* in our minds. Such mental images are typically based on $\mathbb{R}$, $\mathbb{R}^2$, or $\mathbb{R}^3$, which are highly intuitive for humans. Urysohn's lemma and the theorems induced by it establish a crucial connection between these topological intuitions and real spaces.

Meanwhile, I would like to note that “separation” consists of two parts: (1) What objects are we trying to separate? (2) How do we define the separation of these objects?

In the previous section, the answer to (2) always involved *open-set separators*. However, Urysohn demonstrated that in certain cases, this separation can be achieved via *continuous real-valued functions*.

See following lemma and understand how the *separator* is changed from open boundary.

Lemma 17.1. Urysohn's Lemma

Let X be a *normal* space and $A, B \subseteq X$ be disjoint closed sets. Then there exists a continuous function

$$f : X \rightarrow [0, 1]$$

such that $f|_A \equiv 0$ and $f|_B \equiv 1$.

Claim 17.2. Big Picture of Proof

If X is metrizable space (implies normal 15.8) then the separator function will be

$$f(x) = \frac{d(x, A)}{d(x, A) + d(x, B)}$$

However the main result of Urysohn's Lemma states that it is also possible for general normal spaces. **Where can we find the numbers in topological setting?** There are some non-trivial technique using indexing. First, we define *dyadic rationals* by

$$\mathbb{D} = \left\{ \frac{k}{2^n} \mid k, n \in \mathbb{N} \right\} \cap [0, 1]$$

We'll use this set as *index set* and construct a family of open sets $\{U_r\}_{r \in \mathbb{D}}$ such that satisfy

$$r < s \implies A \subseteq U_r \subseteq \overline{U_r} \subseteq U_s \subseteq X \setminus B \quad (\star)$$

From this definition, we finally define

$$f(x) = \inf\{r \in \mathbb{D} \mid x \in U_r\}$$

This is continuous and $f|_A \equiv 0, f|_B \equiv 1$.

Proof. Let $U_1 = X \setminus B$ which is open in X . Then, $A \subseteq U_1$ so by 15.12, we can find an open set U_0 such that

$$A \subseteq U_0 \subseteq \overline{U_0} \subseteq U_1 = X \setminus B$$

Now our goal is constructing $\{U_r\}_{r \in \mathbb{D}}$ holds condition $(\star)$. We'll construct U_r inductively with respect to n . Suppose that if $n \leq m$, we already have

$$\left\{ U_\alpha \mid \alpha = \frac{k}{2^n}, k \in \mathbb{N}, U_\alpha \text{ is open} \right\}, \quad \frac{x}{2^n} = r < s = \frac{y}{2^n} \implies \overline{U_r} \subseteq U_s \quad (\text{I.H.})$$

Now we consider construction of

$$\left\{ U_\alpha \mid \alpha = \frac{k}{2^{m+1}}, k \in \mathbb{N}, U_\alpha \text{ is open} \right\} \quad (1)$$

We define

$$\text{For each } t = \frac{2k+1}{2^{m+1}}, \text{ we can find } r = \frac{k}{2^m}, s = \frac{k+1}{2^m} \text{ so } t = \frac{r+s}{2}.$$

However here, we already have $U_r \subseteq \overline{U_r} \subseteq U_s$. Since $\overline{U_r}$ is closed and U_s is open, we can apply 15.12 and find an open set (denote it U_t) such that

$$r < \frac{r+s}{2} = t < s, \quad \overline{U_r} \subseteq U_t \subseteq \overline{U_t} \subseteq U_s$$

This construct (1) successfully, however such (1) satisfy $(\star)$ clearly. With this inductive construction, we construct

$$\{U_r\}_{r \in \mathbb{D}}, \quad U_r \text{ is open in } X, \quad r < s \implies A = U_0 \subseteq U_r \subseteq \overline{U_r} \subseteq U_s \subseteq U_1 = X \setminus B$$

Now as stated above, define

$$f : X \rightarrow [0, 1] \text{ defined by } f : x \mapsto \begin{cases} \inf \underbrace{\{r \in \mathbb{D} \mid x \in U_r\}}_L & L \neq \emptyset \\ 1 & L = \emptyset \end{cases}$$

Then $f|_A \equiv 0$ since $A \subseteq U_0$. Also, $f|_B \equiv 1$ since $U_1 = X \setminus B$, so $L = \emptyset$ in this case always. And $\text{Im}(f) \subseteq [0, 1]$ since $L \subseteq [0, 1]$. One last thing to prove is that the *continuity of f* . Using 5.4, it is sufficient to show that *preimage of subbasis of codomain is open in domain*. The standard subbasis of $[0, 1]$ is *open rays*, such that for $a \in [0, 1]$, $(-\infty, a)$ and (a, ∞) . In this case,

$$\mathcal{S} = \{[0, a), (a, 1] \mid a \in [0, 1]\}$$

However,

$$f^{-1}([0, a)) = \bigcup_{r \in \mathbb{D}, r < a} U_r, \quad f^{-1}((a, 1]) = \bigcup_{s \in \mathbb{D}, s > a} (X \setminus \overline{U_s})$$

So any preimages of subbasis of $[0, 1]$ is open in X , so f is continuous map. $\square$

Remark 17.3

Try to prove the last part of above proof: showing continuity of f . The dense property of $\mathbb{D}$ is used. Also, try to prove that the converse of Urysohn's Lemma also holds. If we try to change our argument for one closed set and one point and continuous functions, we have $T_{3.5}$, Tychonoff space.

Now we prove the main theorem of this section.

Theorem 17.4. Urysohn's Metrization Theorem

Every regular second countable space is metrizable. Furthermore, it embeds homeomorphically into the Hilbert cube

$$H = [0, 1]^\omega = \prod_{i=1}^{\infty} [0, 1]$$

with the metric

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{n=1}^{\infty} \frac{(x_n - y_n)^2}{n^2}}$$

Proof. Our goal is find a nice embedding from X to metrizable space $[0, 1]^\omega$ and then argue that there exists homeomorphism between X with its image of embedding. Then, since metrizable is topological invariant, X is also metrizable.

Since X is countable, let $\mathcal{B} = \{B_n\}_{n=1}^{\infty}$ be a countable basis for X . Let's define

$$\mathcal{P} = \{(n, m) \in \mathbb{N} \times \mathbb{N} \mid \overline{B_n} \subseteq B_m\}$$

Since $\mathbb{N} \times \mathbb{N}$ is countable, $\mathcal{P}$ also countable so we can denote $\mathcal{P} = \{(n_k, m_k)\}_{k \in \mathbb{N}}$. Here, for each $(n_k, m_k) \in \mathcal{P}$, we can find 2 closed sets: $\overline{B_{n_k}}$ and $X \setminus B_{m_k}$. Remarks that, every second countable regular space is normal by 16.4. So X is normal and we can apply Urysohn's Lemma to $\overline{B_{n_k}}$ and $X \setminus B_{m_k}$.

$$g_k : X \rightarrow [0, 1], \quad g_k|_{\overline{B_{n_k}}} \equiv 0, \quad g_k|_{X \setminus B_{m_k}} \equiv 1$$

Now we have the family $\{g_k\}_{k \in \mathbb{N}}$ of continuous functions from X to $[0, 1]$ that separates closed sets. We first show that X is $T_{3.5}$ space with this construction of $\{g_k\}$. Suppose that $x \in X$ and C be a closed set of X such that $x \notin C$. Then $x \in X \setminus C$ which is open, by 15.6 we can find nice gap V between x and $X \setminus C$ means that

$$x \in V \subseteq \overline{V} \subseteq X \setminus C$$

However, since $\mathcal{B}$ is a basis of X , we can find $B_m \in \mathcal{B}$ such that

$$x \in B_m \subseteq V \subseteq \overline{V} \subseteq X \setminus C$$

Also by 15.6, we can find *nice gap* W between x and B_m means that

$$x \in W \subseteq \overline{W} \subseteq B_m \subseteq V \subseteq \overline{V} \subseteq X \setminus C$$

Finally we find a basis B_n such that $x \in B_n \subseteq W$. It implies that $\overline{B_n} \subseteq \overline{W}$, so we denote

$$x \in \overline{B_n} \subseteq \overline{W} \subseteq B_m \subseteq X \setminus C$$

Now, $(n, m) \in \mathcal{P}$ so they have natural number index (denote k). Then,

$$g_k|_{\overline{B_n}} \equiv 0, \quad g_k|_{X \setminus B_m} \equiv 1, \quad g_k \text{ is continuous}$$

Since $x \in B_n$ and $C \subseteq X \setminus B_m$, we can rewrite

$$g_k(x) = 0, \quad g_k|_C \equiv 1, \quad g_k \text{ is continuous}$$

This separates x and closed set C not containing x via continuous real function, so X is $T_{3.5}$. Now we'll embed X into $[0, 1]^\omega$ by following function $F : X \rightarrow [0, 1]^\omega$

$$F : x \mapsto (g_k(x))_{k \in \mathbb{N}}$$

Then, F is injective. For $x \neq y \in X$, since $\{y\}$ is singleton so closed in X by T_1 axiom. It implies that there exists g_k such that $g_k(x) = 0, g_k(y) = 1$. So $F(x), F(y)$ are different in at least one index k .

Since F is mapping onto product space $[0, 1]^\omega$ and continuous since it is indexwise continuous. Means that each g_k continuous so F is continuous by 6.18.

Finally, F is an open map. Let $x \in X$ and choose any neighborhood U of x in X . Then $X \setminus U$ is closed and $x \notin X \setminus U$, we can find g_k such that $g_k(x) = 0, g_k(X \setminus U) = \{1\}$. Let's define

$$W = F(X) \cap \pi_k^{-1}([0, 1/2))$$

This is open in $F(X)$, $W \subseteq F(U)$ and contains $F(x)$ in $[0, 1]^\omega$. So, $F(U)$ is open in $F(X)$ means that F is an open map.

Injective, continuous, open map forms a homeomorphism from X to its image under F , $F(X)$. So, $X \cong F(X)$ implies that X is metrizable. $\square$

Remark 17.5

1. Every separable metrizable spaces can embedded into Hilbert Cube.
2. Since local Euclidean implies regularity, every topological manifold is metrizable.
3. Every metrizable space is T_3 and every separable metrizable space is second countable.

Remark 17.6. My Intuitions

In this memo, I outline my intuitive understanding of the proof of Urysohn's Metrization Theorem.

The theorem fundamentally states that if a topological space possesses a sufficiently fine “resolution” (second countability) and allows for spatial separation (regularity), we can formally define a distance between its fundamental units, or “pixels.”

We can conceptualize the elements of the space as these pixels, which are locally captured by a countable basis. Provided we have this structural resolution, for every topological “gap” between disjoint closed sets—or clusters of pixels—we can assign a dedicated “dimension.” This dimension acts as a new coordinate axis specifically designed to separate these objects. By assigning these axes, we construct a mapping that embeds the space into a metrizable space (the Hilbert cube).

I studied this part with Lecture note by Hyungryul Baik, KAIST [1].

18 Tietze Extension Theorem

In mathematics, we often study the ‘extension’ like *field extension*. When you softly study Riemann Conjecture, you might also find the word extension. It’s like a dual concept of ‘embeddings’. We’ll *expand* our domain of continuous function in some condition. Actually, the *Tietze Extension Theorem* is one of the most powerful topological theorem we’ll often use further. Also It characterize normality: X is normal if and only if it has the Tietze extension property.

We first prove the key lemma of Tietze extension theorem.

Lemma 18.1. Third Trick

Let X be normal and $A \subseteq X$ be closed. For $h : A \rightarrow [-M, M]$ continuous function, there exists a continuous function $g : X \rightarrow [-M/3, M/3]$ such that

$$|h(x) - g(x)| \leq \frac{2M}{3}, \quad \forall x \in A$$

Proof. We define following two region:

$$A^- = h^{-1}([-M, -M/3]) \quad A^+ = h^{-1}([M/3, M])$$

Since $A^- \cap A^+ = h^{-1}(\emptyset) = \emptyset$, A^- , A^+ are disjoint. Preimage of closed set under continuous map is closed, so A^- , A^+ is closed in A . Moreover since A is closed in X , we deduce that A^- , A^+ are disjoint closed set of X . By Urysohn’s lemma 17.1, we can find $g : X \rightarrow [-M/3, M/3]$ (Since the choice of domain range is arbitrary $[a, b]$) such that $g|_{A^-} \equiv -M/3$ and $g|_{A^+} \equiv M/3$. Then,

$$|h(x) - g(x)|_{x \in A} = \begin{cases} |[-M, -M/3] - (-M/3)| \leq 2M/3 & x \in A^- \\ |[M/3, M] - (M/3)| \leq 2M/3 & x \in A^+ \\ |[-M/3, M/3] - [-M/3, M/3]| \leq 2M/3 & x \in A \setminus (A^- \cup A^+) \end{cases}$$

implies that

$$|h(x) - g(x)|_{x \in A} \leq 2M/3$$

□

Now we're ready to prove main statement of this section.

Theorem 18.2. Tietze Extension Theorem

Let X be a normal space and $A \subseteq X$ be a closed subset.

1. If $f : A \rightarrow [a, b]$ is continuous then there exists continuous $F : X \rightarrow [a, b]$ with $F|_A = f$.
2. If $f : A \rightarrow \mathbb{R}$ is continuous then there exists continuous $F : X \rightarrow \mathbb{R}$ with $F|_A = f$.

Proof. For (1), it is sufficient to show for $f : A \rightarrow [-1, 1]$. (Since the choice is arbitrary) We'll do inductive construction of our target function. Begin with

$$h_0 = f : A \rightarrow [-1, 1]$$

By 18.1, we can find

$$g_1 : X \rightarrow [-1/3, 1/3], \quad \|h_0 - g_1\|_A \leq 2/3$$

So, we can construct

$$h_1 = (h_0 - g_1)|_A : A \rightarrow [-2/3, 2/3]$$

Inductively doing this:

$$h_n = (h_0 - g_1 - g_2 - \cdots - g_n)|_A : A \rightarrow [-(2/3)^n, (2/3)^n]$$

So we can denote it into following way.

$$\|h_0 - \sum_{i=1}^n g_i\|_A \leq \left(\frac{2}{3}\right)^n$$

Let's denote the partial summation of functional series be:

$$S_n = \sum_{i=1}^n g_i$$

However, it is uniformly Cauchy:

$$\|g_n + g_{n+1} + \cdots + g_m\|_X \leq \sum_{i=n}^m \left(\frac{1}{3}\right)^i \rightarrow 0, \quad n, m \rightarrow \infty$$

Hence uniformly converges:

$$\sum_{i=1}^n g_i = S_n \Rightarrow F = \sum_{i=1}^{\infty} g_i$$

Now we claim that F is our target function(Tietze extension function). First, each g_i is continuous since they are from 17.1, and uniform continuity preserves functionwise

continuity so F is continuous. And domain of F is X , also the codomain of F is $[-1, 1]$ since

$$|F(x)| \leq \sum_{i=1}^{\infty} |g_i(x)| \leq \sum_{i=1}^{\infty} \left(\frac{1}{3}\right)^i \leq 1$$

Finally, $F|_A \equiv f$ since

$$\|f - S_n\|_A \leq \left(\frac{2}{3}\right)^n, \quad S_n \rightrightarrows F \implies F|_A \equiv f$$

Now let's prove the second part of the theorem. First,

$$\mathbb{R} \cong (-1, 1) \text{ by } \varphi : t \mapsto \frac{t}{1 + |t|}$$

Then for $f : A \rightarrow \mathbb{R}$, we can construct $\varphi \circ f : A \rightarrow (-1, 1)$. By part (1), we find

$$G : X \rightarrow [-1, 1], \quad G|_A \equiv \varphi \circ f$$

One natural way is obtaining f -extension by $\varphi^{-1} \circ G$. However, if G has $-1, 1$ as its image, φ^{-1} is not defined. So we must modify G properly, let

$$C = G^{-1}(\{-1, 1\}) \implies \text{closed in } X$$

since $\{-1, 1\}$ is discrete set in Hausdorff space so closed, and its preimage under continuous function G also closed in X . And since $G(A)$ does not contain $-1, 1$, C and A are disjoint closed sets in X . So the Urysohn's Lemma 17.1 again implies that there exists

$$\lambda : X \rightarrow [0, 1], \quad \lambda|_C \equiv 0, \quad \lambda|_A \equiv 1$$

Let's construct

$$\bar{G} = \lambda \cdot G$$

We obtain $\bar{G}$ is continuous, and

$$\bar{G}|_A = \lambda(A) \cdot G|_A = G|_A = \varphi \circ f$$

Finally,

$$G(x) = \pm 1 \implies \lambda(x) = 0 \implies \bar{G}(x) = 0$$

So,

$$\bar{G}(X) \subseteq (-1, 1)$$

Now we can well-define

$$F = \varphi^{-1} \circ \bar{G} : X \rightarrow \mathbb{R}, F|_A = f$$

□

Here I append some remarks from [1].

Remark 18.3

If Tietze extension property holds then X is normal.

Proof. Let A, B are disjoint open sets in X . We construct $f : A \rightarrow 0$ and $g : B \rightarrow 1$. Since constant function is continuous, they are continuous in A, B respectively. By 5.8, we can construct

$$h : A \cup B \rightarrow [0, 1], \quad h|_A \equiv f, \quad h|_B \equiv g, \quad h \text{ is continuous}$$

So we can apply Tietze extension theorem:

$$F : X \rightarrow [0, 1], \quad \text{continuous}, \quad F|_A \equiv 0, F|_B \equiv 1$$

This can be viewed as Urysohn separation function in 17.1. Now simply, $F^{-1}([0, 1/2))$ and $F^{-1}((1/2, 1])$ forms open separator of A, B in X . Implies that X is normal. $\square$

Remark 18.4

Closedness of domain is matter: For example, $f(x) = \sin(1/x)$ on $(0, 1]$ has no continuous extension to $[0, 1]$.

19 Homotopy

Now we're ready to dive into 'real' (not $\mathbb{R}$) topology. From now on, we dealt very nice abstraction of drawable spaces via main object path and homotopy. First, we define general notion of homotopy first.

Definition 19.1. Homotopy

Let $f, f' : X \rightarrow Y$ be continuous maps. We say that f is *homotopic* to f' (denoted by $f \simeq f'$) if there exists

$$F : X \times [0, 1] \rightarrow Y, \quad F \text{ is continuous, } \quad F|_{X \times \{0\}} \equiv f, \quad F|_{X \times \{1\}} \equiv f'$$

Such F is called *homotopy* between f and f' . If f is homotopic to constant map, f is called *nul-homotopic*.

Homotopy can be understand as *Continuous(time) change of continuous function*. i.e., We often call $[0, 1]$ as *time axis*. Through out the time going, our function f changed continuously to f' .

Proposition 19.2. Homotopic is Equivalence Relation

The relation *homotopic* ($\simeq$) is equivalence relation. i.e.,

- $f \simeq f$
- $f \simeq f' \iff f' \simeq f$
- $f \simeq g, g \simeq h \implies f \simeq h.$

So we can define equivalence classes under this relation, which is called *homotopy classes*.

Now we define very natural, intuitive notion of drawing *line, path* on space X . We call that

$$\gamma : [0, 1] \rightarrow X, \quad \gamma \text{ is continuous} \implies \gamma \text{ is a path on } X$$

And naturally says $\gamma(0)$ is the *initial point* of γ , and $\gamma(1)$ is the *end point* of γ . Since this notion is more intuitive, useful, and important specific case of 19.1, we specialize them.

Definition 19.3. Path Homotopy

Let $\gamma, \gamma' : [0, 1] \rightarrow X$ be paths. We say that they are *path homotopic relative to end points* (denoted by $\simeq_p$) if

$$\gamma(0) = \gamma'(0), \quad \gamma(1) = \gamma'(1)$$

and there exists

$$F : [0, 1] \times [0, 1] \rightarrow X, \quad F \text{ is continuous, } \quad F|_{[0,1] \times \{0\}} \equiv \gamma, \quad F|_{[0,1] \times \{1\}} \equiv \gamma'$$

You can find a few nice examples in [2] page 325-326. However, we can generalize the notion of ‘concatenation’ of two paths on space as following.

Definition 19.4. Path Product

Let $\gamma, \delta : [0, 1] \rightarrow X$ be paths on X such that $\gamma(1) = \delta(0)$. Then we can construct following path:

$$\gamma * \delta : [0, 1] \rightarrow X, \quad t \mapsto \begin{cases} \gamma(2t) & t \in [0, 1/2] \\ \delta(2t - 1) & t \in [1/2, 1] \end{cases}$$

This is well define based on pasting lemma 5.8.

We can just imagine that 2 times faster move through γ then δ . By the way, as we noted at 19.2, we can find *path homotopic classes* of path γ in X :

$$[\gamma] = \{\delta \mid \delta \simeq_p \gamma, \delta \text{ is any path in } X\}$$

Then suprisingly, we can define the natural binary operation between *path homotopic classes*.

Definition 19.5. Define Binary Operation on Path Homotopic Classes

We can define following binary operation $*$:

$$[\gamma] * [\delta] \mapsto [\gamma * \delta]$$

Then we’ll not prove, but following holds (very naturally)

1. This is well-defined, i.e., $[\gamma] * [\delta] = [\gamma'] * [\delta']$ for $\gamma \simeq_p \gamma', \delta \simeq_p \delta'$.
2. $[\gamma] * ([\delta] * [\eta]) = ([\gamma] * [\delta]) * [\eta]$
3. For $\gamma(0) = x_0, \gamma(1) = x_1$, define e_x be constant map onto x . Then, we obtain

$$[\gamma] * [e_{x_1}] = [e_{x_0}] * [\gamma] = [\gamma]$$

4. We can define $\bar{\gamma}(s) = \gamma(1 - s)$, the reverse path. Then,

$$[\gamma] * [\bar{\gamma}] = [e_{x_0}], [\bar{\gamma}] * [\gamma] = [e_{x_1}]$$

5. Let γ be a path and suppose that $0 \leq a_0 < a_1 < \cdots < a_n = 1$ and $\gamma_i : I \rightarrow X$ induced by $\gamma(a_{i-1})$ to $\gamma(a_i)$. Then,

$$[\gamma] = [\gamma_1] * \cdots * [\gamma_m]$$

20 The Fundamental Groups

At this moment, one beautiful combination between Algebraic notions and Topological notions. One interesting story when I studied this parts, my Topology professor taught us about ‘free groups’ and in the following Algebra class, Algebra professor taught us about ‘fundamental groups’.

We define bunch of important definitions here.

Definition 20.1. The Fundamental Groups

If some path has same initial/end point x_0 , we specialize them to call a *loop based at x_0* . The set of path homotopy classes of loops based at x_0 is denoted by $\pi_1(X, x_0)$. With binary operation defined in 19.5, $\pi_1(X, x_0)$ is group and it is called *the fundamental group of X relative to the base point x_0* .

First natural curious is that, what if we change the ‘base point x_0 ’ on the same space X ? Once there exists a path α between x_0 and x_1 in X , we can construct following natural map.

$$\hat{\alpha} : \pi_1(X, x_0) \rightarrow \pi_1(X, x_1) \text{ defined by } [\gamma] \mapsto [\bar{\alpha}] * [\gamma] * [\alpha]$$

Since the drawing of image of this is obvious, I omitted here. We can observe following property.

Proposition 20.2. Isomorphism Between Fundamental Groups

Let α be a path from x_0 to x_1 in X . $\hat{\alpha}$ is group isomorphism. i.e.,

$$\pi_1(X, x_0) \cong \pi_1(X, x_1)$$

Proof. It is easy to show that this is homomorphism:

$$[\gamma] * [\delta] = [\gamma * \delta] \mapsto [\bar{\alpha}] * [\gamma * \delta] * [\alpha]$$

By the way,

$$\begin{aligned} & [\bar{\alpha}] * [\gamma] * [\alpha] * [\bar{\alpha}] * [\delta] * [\alpha] \\ &= [\bar{\alpha}] * [\gamma] * [e_{x_0}] * [\delta] * [\alpha] \\ &= [\bar{\alpha}] * [\gamma * \delta] * [\alpha] \end{aligned}$$

To show that $\hat{\alpha}$ is bijection, it is enough to show that there exists inverse homomorphism. Define

$$(\hat{\alpha})^{-1} : \pi_1(X, x_1) \rightarrow \pi_1(X, x_0) \text{ defined by } [\delta] \mapsto [\alpha] * [\delta] * [\bar{\alpha}]$$

Hence, $\hat{\alpha}$ is isomorphism. □

Corollary 20.3. Fundamental Group in Path Connected Space

If X is path-connected, $\pi_1(X, x_0)$ does not depend on choice of $x_0 \in X$. i.e., they are all isomorphic.

We'll mainly talk about path connected spaces. The reason is similar why we usually talk about connected graphs in Graph Theory. Since each disconnected components does not have any channel to communicate, it is natural. And note that, **isomorphism between $\pi_1(X, x_0) \xrightarrow{\sim} \pi_1(X, x_1)$ depends on choice of path between them.** However, we can observe following:

$$\pi_1(X, x_0) \xrightarrow{\sim} \pi_1(X, x_1) \text{ independent to path} \iff \pi_1(X, x_0) \text{ is Abelian}$$

Proof. Let $[\gamma], [\delta] \in \pi_1(X, x_0)$. Suppose that α is a path from x_0 to x_1 . Then we can construct another path $[\delta * \alpha]$. Since the assumption,

$$[\bar{\alpha}] * [\gamma] * [\alpha] = [\overline{\delta * \alpha}] * [\gamma] * [\delta * \alpha]$$

This implies that

$$[\gamma] = [\bar{\delta}] * [\gamma] * [\delta] \implies [\delta] * [\gamma] = [\gamma] * [\delta]$$

And converse is clear. □

Definition 20.4. Simply Connected Space

Suppose that X is path-connected space. We say it is *simply connected* when

$$\pi_1(X, x_0) = \{[e_{x_0}]\}$$

i.e., trivial group. Intuitively, our space X is very nice as there is not any 'hole'.

Now the second natural curious is that, what if we change the 'space X ' via continuous maps or homeomorphisms?

Definition 20.5. Homomorphism Induced by Continuous Map

Let $f : (X, x_0) \rightarrow (Y, y_0)$ be a continuous mapping. We can define

$$f_* : \pi_1(X, x_0) \rightarrow \pi_1(Y, y_0) \text{ defined by } [\gamma] \mapsto [f \circ \gamma]$$

This f_* is called the *homomorphism induced by h relative to the base point x_0 .*

It is not hard to see that f_* is well-defined homomorphism. Note that, as very clear reason, **If f is homeomorphism then f_* is group isomorphism.**

21 Universal Covering Theory - Introduction

Many topological spaces we encounter in practice are highly complex—riddled with holes, twisted structures, and intricate walls. Imagining such spaces, especially in higher dimensions, can be a daunting task. On the other hand, simply connected spaces, such as $\mathbb{R}^n$, are completely flat and mathematically "clean," making them exceptionally natural and easy to work with. Fortunately, these simple spaces often serve as the fundamental building blocks for more complicated ones via quotient topologies. We have already seen a prime example of this: constructing a torus by gluing the edges of a simple, well-behaved square. Now, we want to take this idea further. Our goal is to reverse this process—to systematically "unfold" or "unroll" any twisted, complex space into a pure, simply connected space.

Here I write up the entire big picture of this section here. We'll state that under some condition of B , we can 'maximally' unfold it into simply connected space (called universal cover) $\tilde{B}$. However, above some condition containing path-connected, hence B has unique first fundamental group $\pi_1(B)$. Try to view that group acts on $\tilde{B}$. Then finally, it determines the folded structure of B from $\tilde{B}$, means that

$$B \cong \tilde{B}/\pi_1(B)$$

Hence, for such good B , it is enough to analysis fundamental group $\pi_1(B)$. Via them, we can fully understand the topological structures and folded structure of B from $\tilde{B}$.

Now I give very explicit example in this page. Try to view following complex function as topological manner:

$$p : \mathbb{R} \rightarrow S^1 \cong \text{unit circle in } \mathbb{C}, \text{ defined by } t \mapsto e^{2\pi i t}$$

The fundamental group $\pi_1(S^1) \cong \mathbb{Z}$. It can be imagined as n -times loop in S^1 . Suppose that we start from 0 in $\mathbb{R}$. Here, n -times loop can be viewed as move 0 to n in $\mathbb{R}$. i.e., stairup n -times in $\mathbb{R} \implies n$ -loop in S^1 . Hence, each n -loop acts on $\mathbb{R}$ by $x \mapsto x + n$. And also we obtain $S^1 \cong \mathbb{R}/\mathbb{Z}$. Now we 'fold' simply connected space $\mathbb{R}$ with respect to $\mathbb{Z}$, then it construct S^1 circle.

22 Covering Spaces and Lifting Theorems

Now we begin the main theory. First, we define the *covering map*. In basic, it is a natural-map from **more unfolded space into folded target space**. However, in real world also, the fold operation makes some symmetric structures like Decalcomani. Try to view the notion of covering map and evenly covered as Decalcomani in general space.

Remark 22.1. My Intuition

I think we can understand all of these story as *parallel universe theory*. B is our space, and E is God's space(Entire Universe). There are many Homeomorphic parallel universes (sheets). If we do some action $f : Y \rightarrow B$, then it effect *some other parallel universe* on E . In this view, the name p can be viewed as *projection*.

Definition 22.2. Evenly Covered, Covering Map

Suppose that $p : E \rightarrow B$ is continuous surjection. For open set U in B , it is called *evenly covered by p* if

(1)

$$p^{-1}(U) = \bigsqcup_{\alpha \in A} V_{\alpha}$$

means that each V_{α} are pairwise disjoint open sets.

(2) Each V_{α} is homeomorphic to U by $p|_{V_{\alpha}}$

We say that p is *covering map* if for any $b \in B$, we can find evenly covered neighborhood U of b . We'll say that such V_{α} are *sheets* over U .

Then, we can obtain some easy facts once $p : E \rightarrow B$ is covering map.

Proposition 22.3. Property of Covering Map

Suppose that $p : E \rightarrow B$ is covering map.

- (1) p is local homeomorphism.
- (2) p is open map.
- (3) Fibers of single point is discrete.

Proof. (1), (3) is trivial. Suppose that $V \subseteq E$ is open set. Then for $\forall e \in E$, we can find sheet $V_e \subseteq E$ such that containing e . However, $p|_{V_e}$ is homeomorphism, hence $p|_{V_e}(V \cap V_e)$ is open in B . $p(V) = \bigcup_{e \in V} p|_{V_e}(V \cap V_e)$, arbitrary union of open sets hence open in B . $\square$

Definition 22.4. Deck Transformation

Suppose that covering map $p : E \rightarrow B$ is given. A homeomorphism $\varphi : E \rightarrow E$ is called *deck transformation* if $p \circ \varphi = p$. The set of all deck transformations forms a group, denoted by $\text{Deck}(E/B)$ or $\text{Aut}(p)$.

Here we observe a few important facts. As we've already said, $\pi_1(S^1) \cong \mathbb{Z}$. However, it is clear that the only deck transformations of $p : \mathbb{R} \rightarrow S^1$ defined by $t \mapsto e^{2\pi i t}$ is $\varphi : t \mapsto t + n$ for any $n \in \mathbb{Z}$. Hence, $\text{Deck}(E/B) = \mathbb{Z} \cong \pi_1(S^1)$. For another example, let's imagine that $p : S^n \rightarrow P\mathbb{R}^n$. Here, $P\mathbb{R}^n$ is all 1-dim vector spaces in $\mathbb{R}^n$. Then we can imagine that all possible lines over $\mathbb{R}^3$ and the outer-sphere(S^3). Any line in $\mathbb{R}^3$ determines 2 symmetric points over the surface of sphere. Hence, any open sets in $P\mathbb{R}^n$ construct the open sets over the surface of S^n and it is symmetrical through the origin. Hence, $p : S^n \rightarrow P\mathbb{R}^n$ is 2-covering map. The only non-trivial deck transformation is $x \mapsto -x$, hence $\text{Deck}(E/B) \cong Z_2 \cong \mathbb{Z}/2\mathbb{Z}$. However, $\pi_1(P\mathbb{R}^n) \cong \mathbb{Z}/2\mathbb{Z}$ is known. These results reveals that deck transformation group is isomorphic to fundamental groups of codomain.

Theorem 22.5. Deck Transformation is Unique up to point

Let $p : E \rightarrow B$ be covering map and E is connected. φ_1, φ_2 both are deck transformation.

$$\exists e \in E, \varphi_1(e) = \varphi_2(e) \implies \varphi_1 = \varphi_2$$

Proof. Define

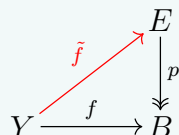
$$S = \{e \in E \mid \varphi_1(e) = \varphi_2(e)\} \neq \emptyset$$

If S is clopen, $S = E$ since E is connected space. First, S is open: suppose that $s \in S \implies \varphi_1(s) = \varphi_2(s) = x$. Then we can find a sheet V over $p(x)$ containing x . Then, $\varphi_1^{-1}(V) \cap \varphi_2^{-1}(V)$ is open neighborhood of s such that φ_1, φ_2 agree. Similarly, S is closed: if $s \in E \setminus S \implies \varphi_1(s) \neq \varphi_2(s)$. Hence, they are one the different sheet V_1, V_2 . Then we can also find $\varphi_1^{-1}(V_1) \cap \varphi_2^{-1}(V_2)$ is open and contained in $E \setminus S$. $\square$

Now we define one more important concept, which is called *lifting*. Up to this point, we understand the concept of covering map as Decalcomani. However, we did nothing yet. To give the meaning here, we just need to draw some picture on our folded space B and unfold it onto E through covering map (fold operation).

Definition 22.6. Lifting

Suppose that $p : E \rightarrow B$ is a covering map. Let $f : Y \rightarrow B$ is continuous map. We say that $\tilde{f} : Y \rightarrow E$ is a *lifting of f by p* if $\tilde{f}$ is continuous and $p \circ \tilde{f} = f$.



See 22.1. From now on, I'll try to explain my understanding about all these notion as this way. Suppose that we do something($f : Y \rightarrow B$) onto our space B . It happens in parallel universes in E . Once we can fix one element of some parallel universe, we can fix our entire f on such universe, which is lifting of our f .

Theorem 22.7. Unique Lifting Theorem

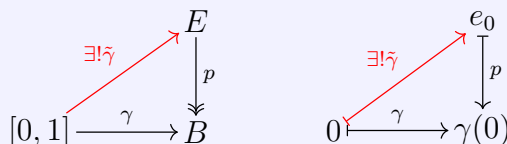
Let $p : E \rightarrow B$ be covering map, Y is connected space and $f : Y \rightarrow B$ is continuous map. Suppose that $\tilde{f}_1, \tilde{f}_2$ are lifting of f by p . If $\tilde{f}_1, \tilde{f}_2$ agree at least one point in E , they are exactly same. Hence lifting of f by p is unique up to choice of one point.

Proof. Do something from 22.5. □

Now, since our goal is deal more intuitive thing which is fundamental group, we'll specify 2 cases using the notion of lifting. First one is path lifting, it can be imagined that we draw some path in our space B , then every parallel spaces in E maybe draw it its own way. Second one is Homotopy lifting. If we continuously change our path in B , every parallel space can do similar thing in E .

Theorem 22.8. Path Lifting Theorem

Let $p : E \rightarrow B$ be a covering map. Suppose that $\gamma : [0, 1] \rightarrow B$ is a path on B . Fix one $e_0 \in E$ such that $p(e_0) = \gamma(0)$. There exists the unique path lifting $\tilde{\gamma} : [0, 1] \rightarrow E$ of γ such that $\tilde{\gamma}(0) = e_0$ and $p \circ \tilde{\gamma} = \gamma$.



Proof. Since p is covering map, for any $\gamma(t) \in B$ where $t \in [0, 1]$, we can find U_t : evenly covered neighborhood of $\gamma(t)$. Then, $\gamma^{-1}(U_t)$ forms open covering of $[0, 1]$. By 12.5, we can find Lebesgue number $\delta > 0$ such that if $S \subseteq [0, 1]$ and $\text{diam}(S) < \delta$ then S entirely contained in some $\gamma^{-1}(U_t)$. Then now, partitioning $[0, 1]$ into $[t_i, t_{i+1}]$ where $t_0 = 0, t_m = 1$ and $t_{i+1} - t_i < \delta$. Now each $[t_i, t_{i+1}] \subseteq \gamma^{-1}(U_t)$ for some t . However, $t_0 = 0$ maps to e_0 in E and $\gamma(0)$ in B . Hence we can choose sheet containing e_0 , then $[t_0, t_1] \rightarrow E$ is uniquely determined when $p \circ \tilde{\gamma} = \gamma$ on $[t_0, t_1]$. Hence we can fix $\tilde{\gamma}(t_1)$. Do this for $[t_1, t_2]$, and so on. By pasting lemma 5.8, we can all concate them, it construct $\tilde{\gamma} : [0, 1] \rightarrow E$ such that $\tilde{\gamma}(0) = e_0$ and $p \circ \tilde{\gamma} = \gamma$. $\square$

Theorem 22.9. Homotopy Lifting Theorem

Let $p : E \rightarrow B$ be a covering map. Suppose that $H : [0, 1] \times [0, 1] \rightarrow B$ is a path homotopy on B . Once initial path lifting $\tilde{H}|_{[0,1] \times \{0\}}$ is fixed on E , there exists a unique lifting $\tilde{H}$ of H in E such that

$$\tilde{H} : [0, 1] \times [0, 1] \rightarrow E, \quad p \circ \tilde{H} = H, \quad \tilde{H}(\cdot, 0) = \text{lift of } H(\cdot, 0)$$

Proof. Do similarly with 22.8 on 2D plane. $\square$

Corollary 22.10. End points of Lifting of Homotopy

Let $p : E \rightarrow B$ be a covering map. 2 paths γ_1, γ_2 are homotopic in B rel end points. Suppose that $\tilde{\gamma}_1, \tilde{\gamma}_2$ are there path lifting by p . Once they share the initial point, then they share the end points: $\tilde{\gamma}_1(1) = \tilde{\gamma}_2(1)$.

Proof. Construct homotopy $H : [0, 1] \times [0, 1] \rightarrow B$ from γ_1 to γ_2 . As $\tilde{\gamma}_1$ is given, fix it as initial path on E , then there exists unique homotopy lifting $\tilde{H}$ by 22.9. Now suppose $\tilde{H}(1, \cdot)$. Since $H(1, \cdot)$ is constant map on $\gamma_1(1) = \gamma_2(1) = b$, $p \circ \tilde{H}(1, \cdot)$ also constant on b . However, $p^{-1}(b)$ is discrete and $\tilde{H}$ is continuous, hence $\tilde{H}(1, \cdot)$ also constant map. It implies that the end points of $\tilde{\gamma}_1, \tilde{\gamma}_2$ are same. $\square$

Corollary 22.11. The Induced Homomorphism Between Fundamental Groups

Once covering map $p : E \rightarrow B$ is given, then the induced homomorphism 20.5

$$p_* : \pi_1(E, e_0) \rightarrow \pi_1(B, b_0)$$

is injective.

Proof. Homomorphism is injective if and only if its kernel is trivial. Suppose that $p_*([\gamma]) = [C_{b_0}]$. By definition, it implies that there exists a path homotopy between $p \circ \gamma \simeq_p C_{b_0}$. See the lifting of C_{b_0} in E . Since $p^{-1}(b_0)$ is discrete and one point e_0 is fixed, lifting of C_{b_0} is C_{e_0} in E . Hence by the homotopy lifting theorem 22.9, lifting of

$p \circ \gamma$ is path homotopic to C_{e_0} . We can specify lifting of $p \circ \gamma$ by γ , hence $\gamma \simeq_p C_{e_0}$, hence $[\gamma] = [C_{e_0}]$ which is identity in $\pi_1(E, e_0)$. $\square$

From now on, we've shown that the existence of lifting only for path and homotopy by direct constructing. One natural question is the existence of lifting of general continuous function $f : Y \rightarrow B$.

Following theorem intuitively means that, lifting exists if and only if Y is unfolded more than E . View $f_*(\pi_1(Y, y_0))$ as complexity of space of Y .

Theorem 22.12. The Lifting Criterion

Let $p : (E, e_0) \rightarrow (B, b_0)$ be a covering map, and $f : (Y, y_0) \rightarrow (B, b_0)$ be continuous map with Y is path-connected and locally path-connected.

Exists unique lift, $\tilde{f} : (Y, y_0) \rightarrow (E, e_0) \iff f_*(\pi_1(Y, y_0)) \leq p_*(\pi_1(E, e_0))$

Proof. ($\Rightarrow$) If such lifting is exists, $p \circ \tilde{f} = f$. Hence, $f_*(\pi_1(Y, y_0))$ is filtered-image of $p_*(\pi_1(E, e_0))$. (Since functorial property: $p \circ \tilde{f} = f \implies p_* \circ \tilde{f}_* = f_*$) Hence

$$f_*(\pi_1(Y, y_0)) \leq p_*(\pi_1(E, e_0))$$

($\Leftarrow$) Here we construct $\tilde{f}$ directly. Choose $y \in Y$, then since Y is path-connected, there exists a path α from y_0 to y . It is projected into B by f :

$$\alpha \mapsto f \circ \alpha, \quad f \circ \alpha(0) = b_0$$

Then we can find its lifting by fixing initial point by e_0 in E .

$$f \circ \alpha \xrightarrow{\text{lift}} \widetilde{f \circ \alpha}, \quad \widetilde{f \circ \alpha}(0) = e_0$$

Then we definitionally define

$$\tilde{f} : y \mapsto \widetilde{f \circ \alpha}(1)$$

Here we need to show that $\tilde{f}$ is well-defined (independent to the choice of α) and valid lifting. Suppose that we choose another y_0 to y path, namely β . We want to show that

$$\widetilde{f \circ \alpha}(1) = \widetilde{f \circ \beta}(1)$$

Here, $\alpha * \bar{\beta}$ forms y_0 loop in Y . Hence by the assumption, $[\alpha * \bar{\beta}] \in \pi_1(Y, y_0)$ implies that

$$[f \circ (\alpha * \bar{\beta})] = p_*([\gamma]), \text{ for some } \gamma \in \pi_1(E, e_0)$$

Hence, lifting of $f \circ (\alpha * \bar{\beta})$ forms loop based at e_0 in E . But lifting of $f \circ (\alpha * \bar{\beta})$ can be viewed as concatenation of both lifts, implies our goal. Showing continuity is not hard. $\square$

Up to these results, we now possesses the tool to compute the fundamental group of S^1 rigorously.

Theorem 22.13. Fundamental Group of S^1

$p : t \mapsto e^{2\pi i \cdot t}$ be our covering map. $\tilde{\gamma}$ is lifting of γ by fixing initial point by $0 \in \mathbb{R}$.

$$\pi_1(S^1, x_0) \cong \mathbb{Z}, \text{ defined by } [\gamma] \mapsto \tilde{\gamma}(1)$$

Proof. This will be nice practice. □

References

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